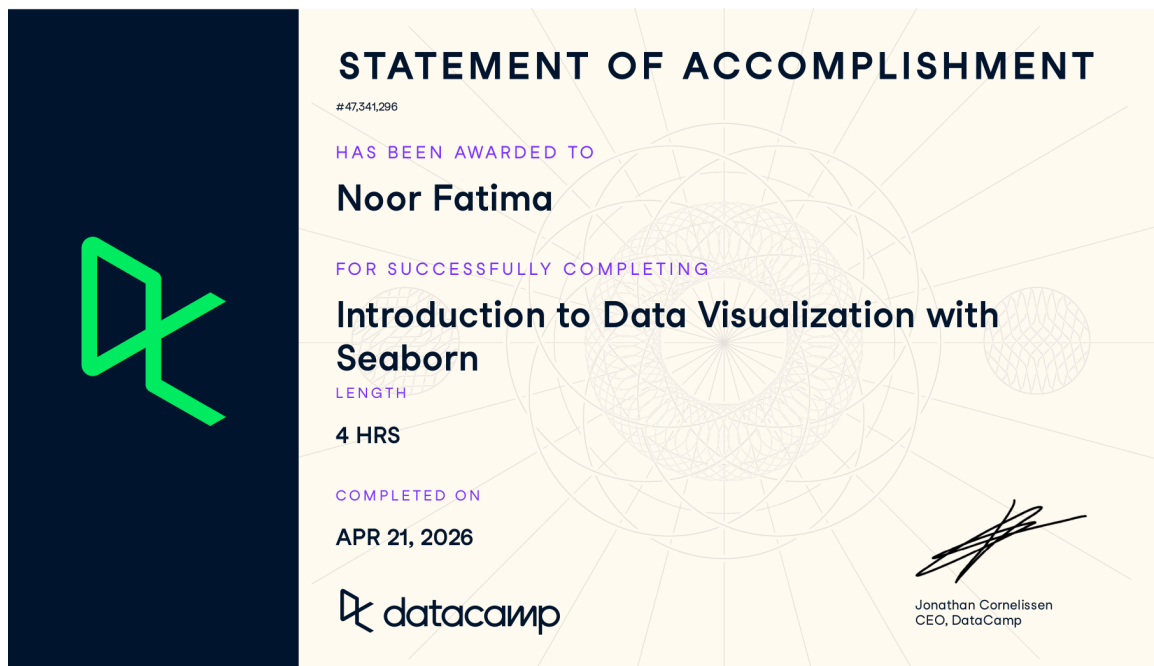


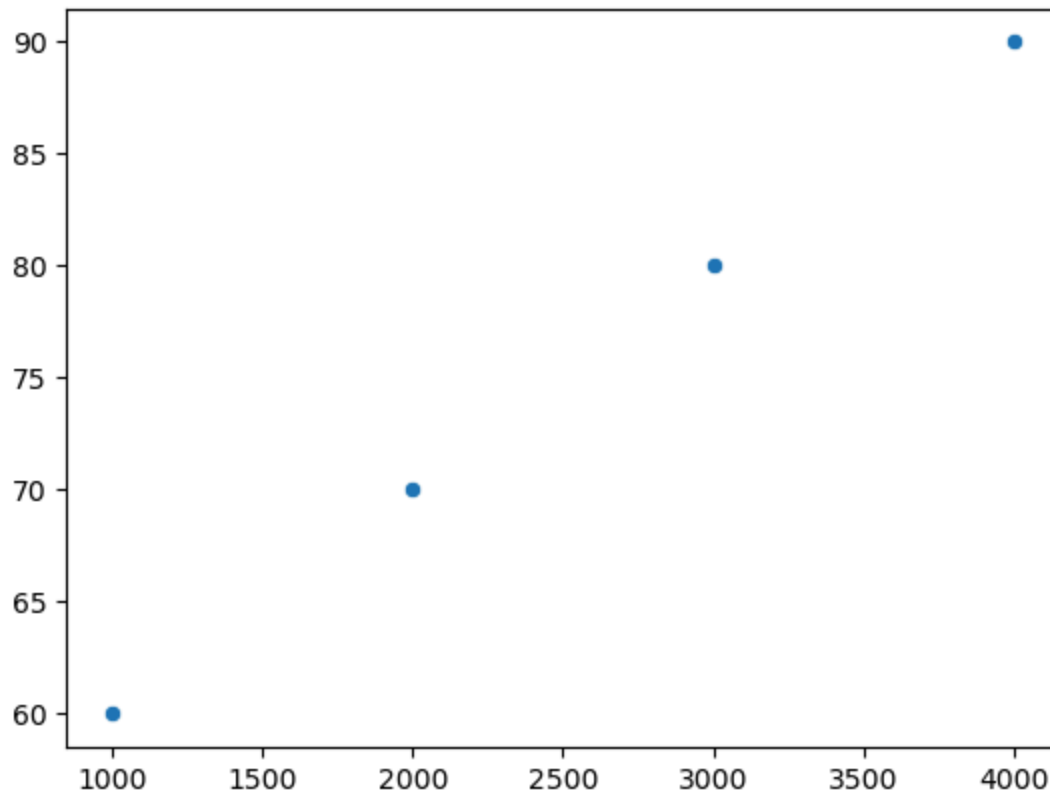


INTRODUCTION TO DATA VISUALIZATION WITH SEABORN:

```
In [1]: import matplotlib.pyplot as plt
import seaborn as sns

gdp = [1000, 2000, 3000, 4000]
percent_literate = [60, 70, 80, 90]

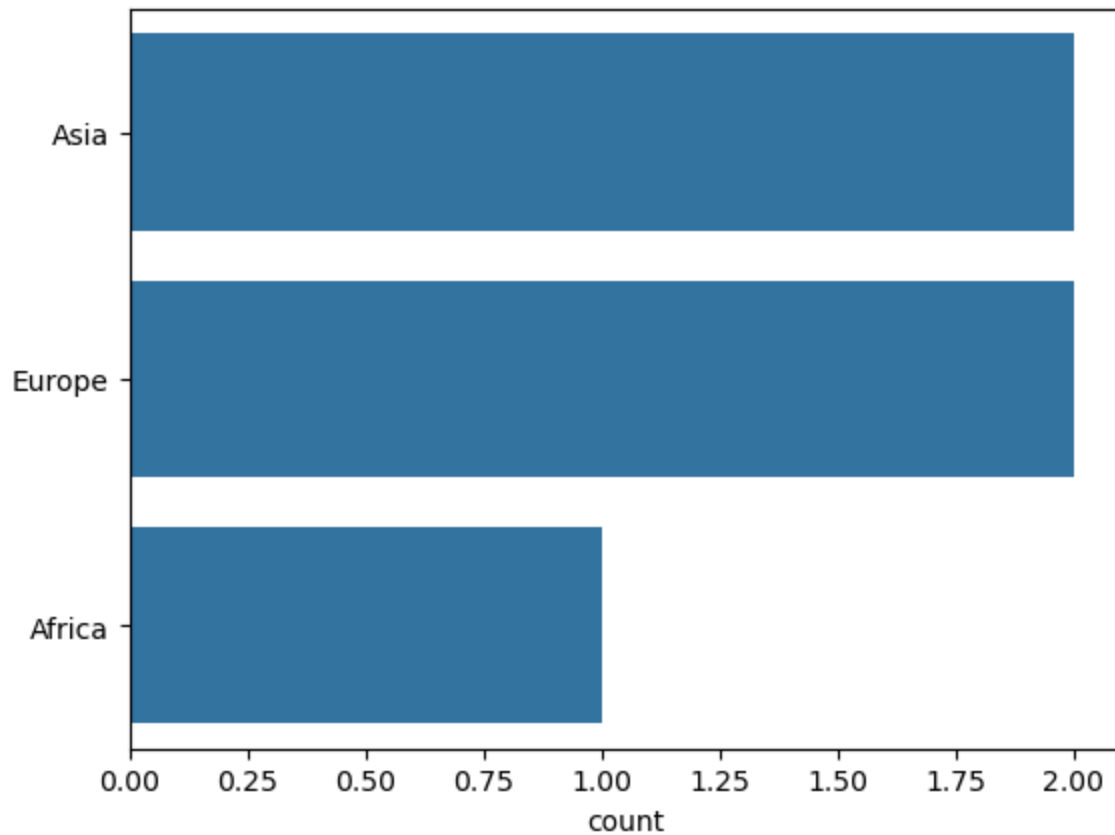
sns.scatterplot(x=gdp, y=percent_literate)
plt.show()
```



```
In [2]: import matplotlib.pyplot as plt
import seaborn as sns

region = ["Asia", "Europe", "Asia", "Africa", "Europe"]

sns.countplot(y=region)
plt.show()
```

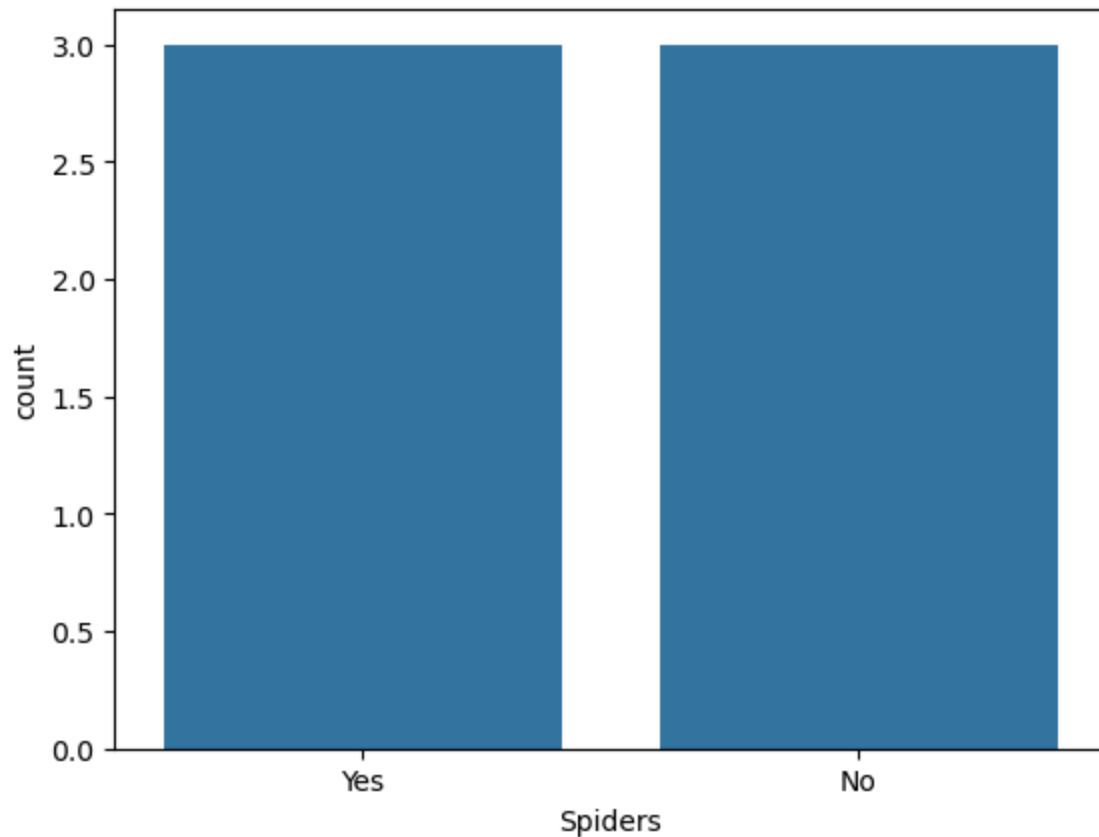


```
In [3]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

data = {
    "Region": ["Asia", "Europe", "Africa", "Asia", "Europe", "Africa"],
    "Spiders": ["Yes", "No", "Yes", "No", "Yes", "No"]
}

df = pd.DataFrame(data)

sns.countplot(x="Spiders", data=df)
plt.show()
```

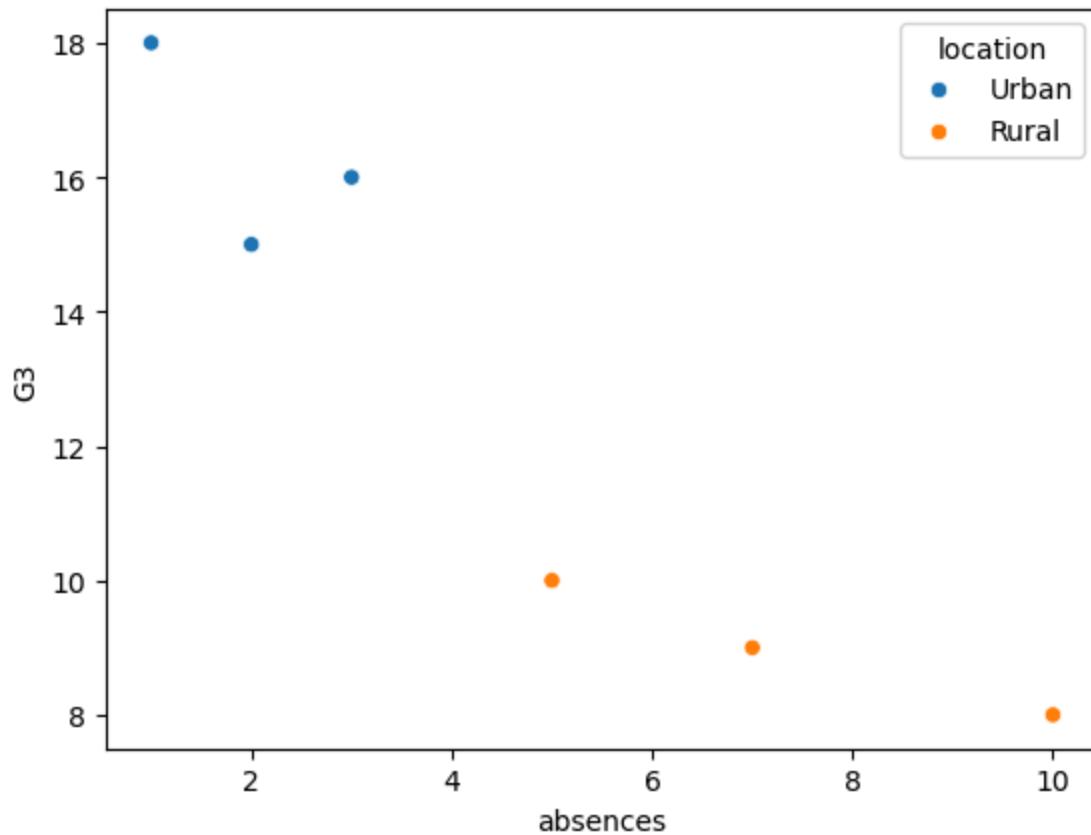


```
In [4]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

student_data = pd.DataFrame({
    "absences": [2, 5, 1, 10, 3, 7],
    "G3": [15, 10, 18, 8, 16, 9],
    "location": ["Urban", "Rural", "Urban", "Rural", "Urban", "Rural"]
})

sns.scatterplot(x="absences", y="G3",
                data=student_data,
                hue="location")

plt.show()
```



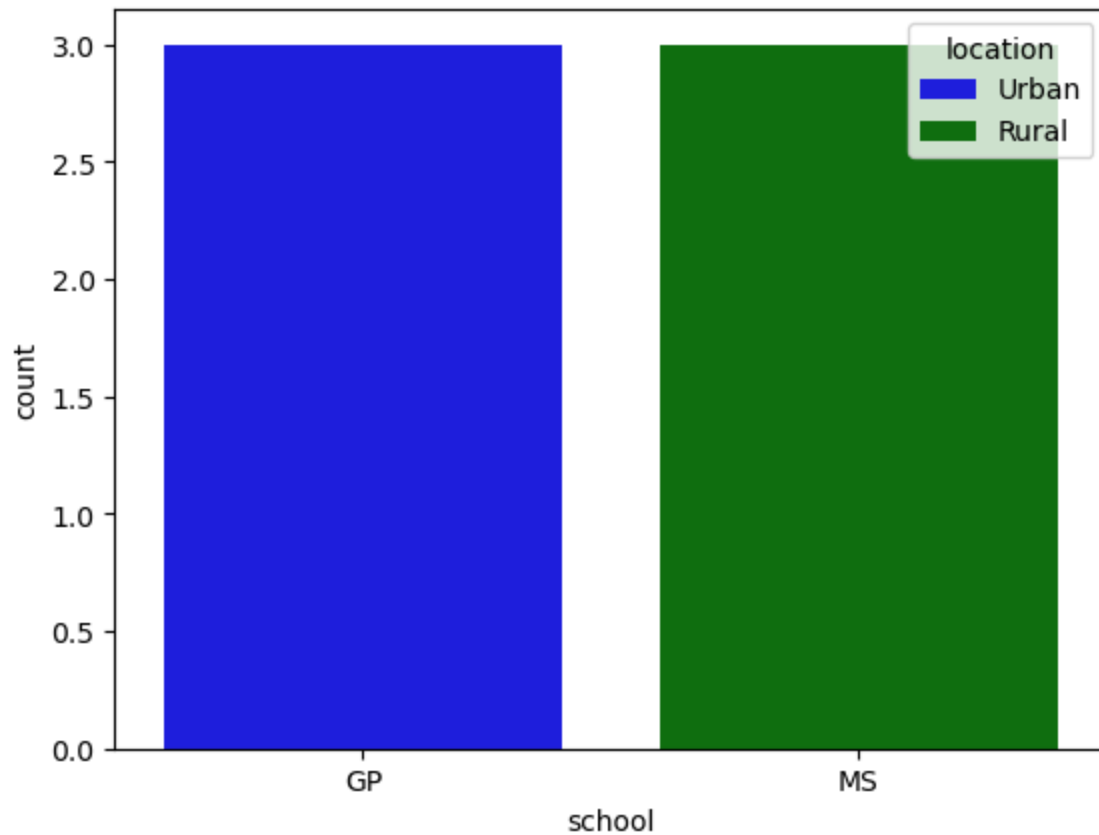
```
In [5]: import matplotlib.pyplot as plt
import seaborn as sns

palette_colors = {"Rural": "green", "Urban": "blue"}

student_data = pd.DataFrame({
    "absences": [2, 5, 1, 10, 3, 7],
    "G3": [15, 10, 18, 8, 16, 9],
    "location": ["Urban", "Rural", "Urban", "Rural", "Urban", "Rural"],
    "school": ["GP", "MS", "GP", "MS", "GP", "MS"]
})

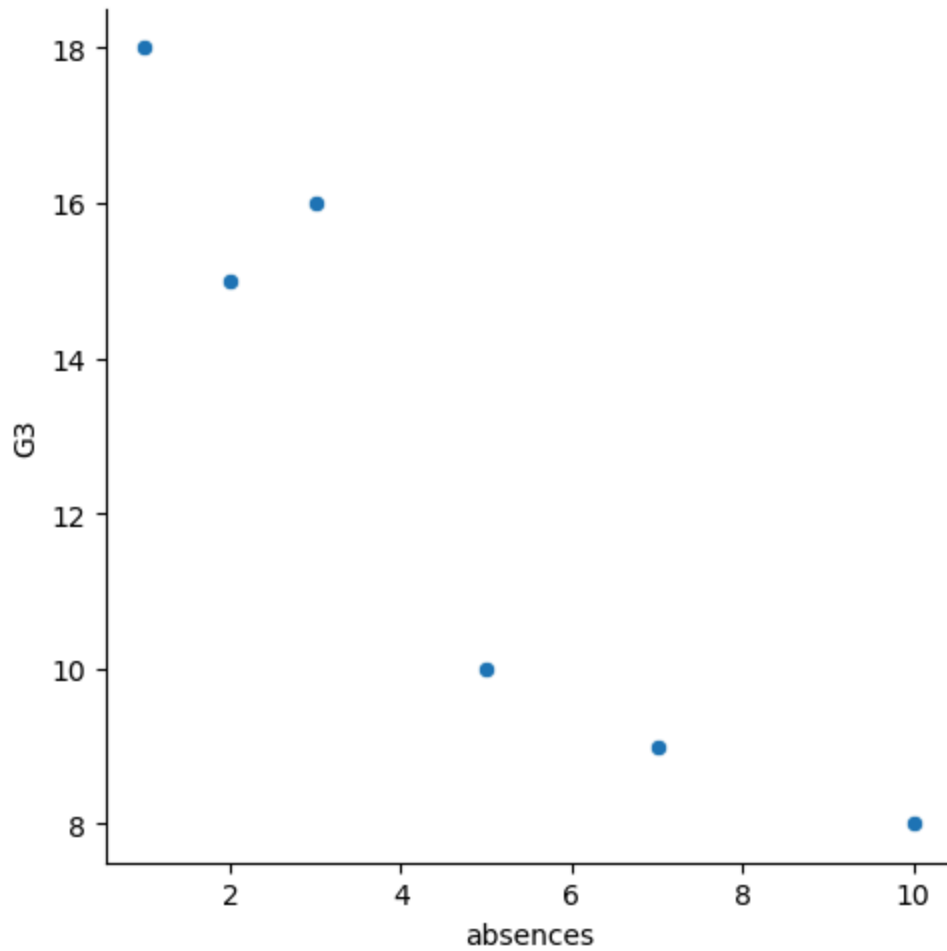
sns.countplot(x="school", data=student_data,
              hue="location",
              palette=palette_colors)

plt.show()
```



INTRODUCTION TO RELATIONAL PLOTS AND SUBPLOTS:

```
In [6]: sns.relplot(x="absences", y="G3",  
                    data=student_data,  
                    kind="scatter")  
  
plt.show()
```

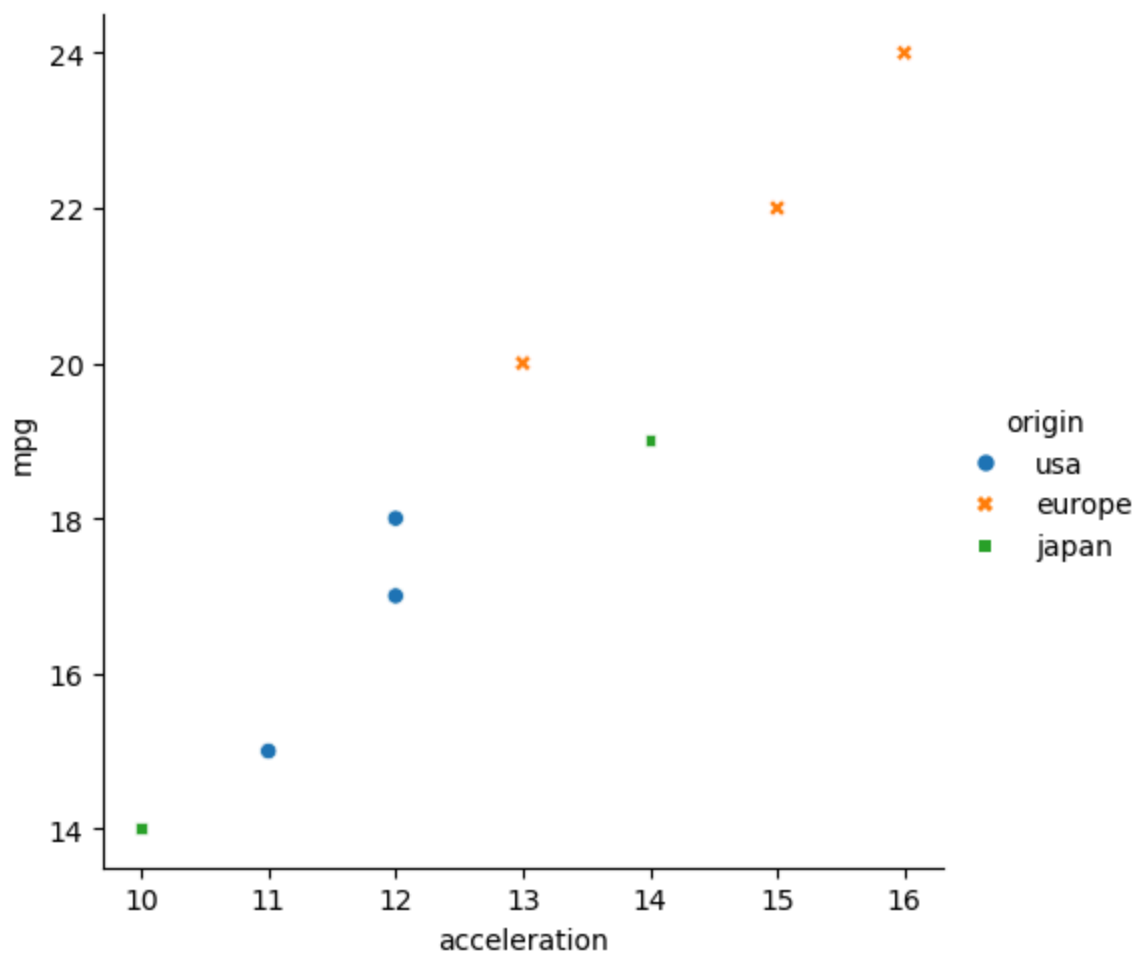


```
In [7]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

mpg = pd.DataFrame({
    "acceleration": [12, 11, 13, 10, 15, 14, 16, 12],
    "mpg": [18, 15, 20, 14, 22, 19, 24, 17],
    "origin": ["usa", "usa", "europe", "japan", "europe", "japan", "europe", "usa"]
})

sns.relplot(x="acceleration", y="mpg",
            data=mpg, kind="scatter",
            style="origin", hue="origin")

plt.show()
```

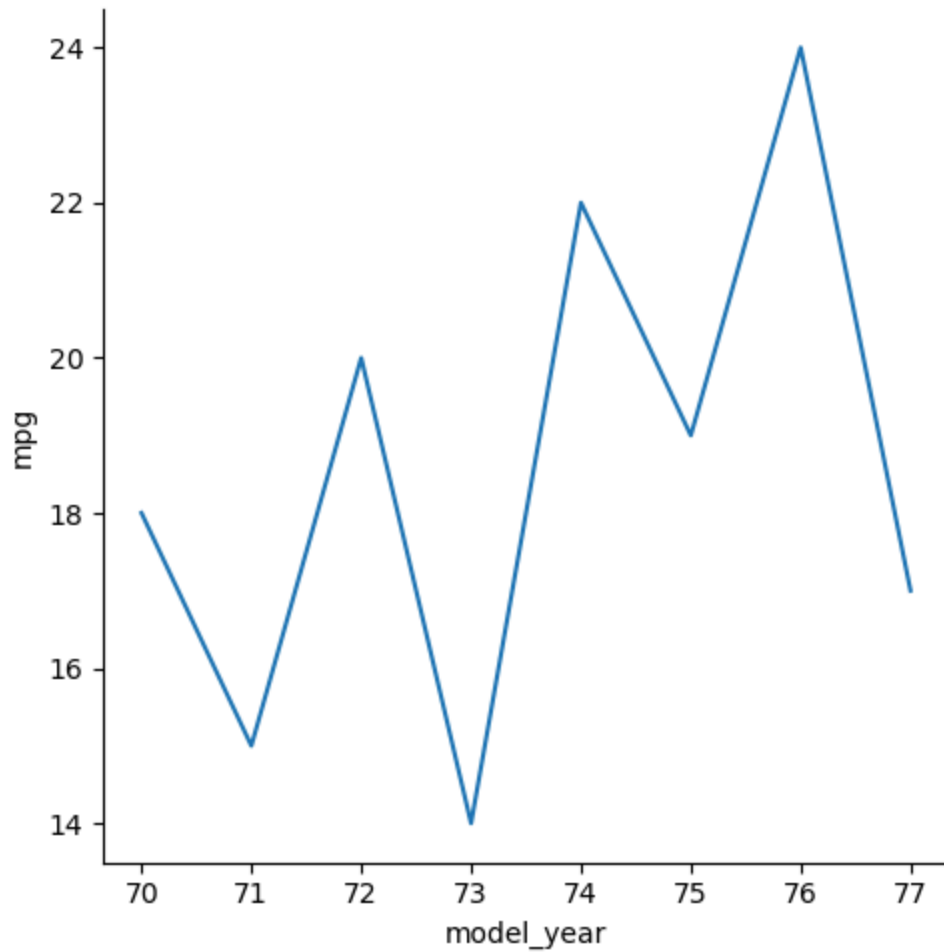


```
In [8]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

mpg = pd.DataFrame({
    "model_year": [70, 71, 72, 73, 74, 75, 76, 77],
    "mpg": [18, 15, 20, 14, 22, 19, 24, 17]
})

sns.relplot(x="model_year", y="mpg",
            data=mpg, kind="line")

plt.show()
```

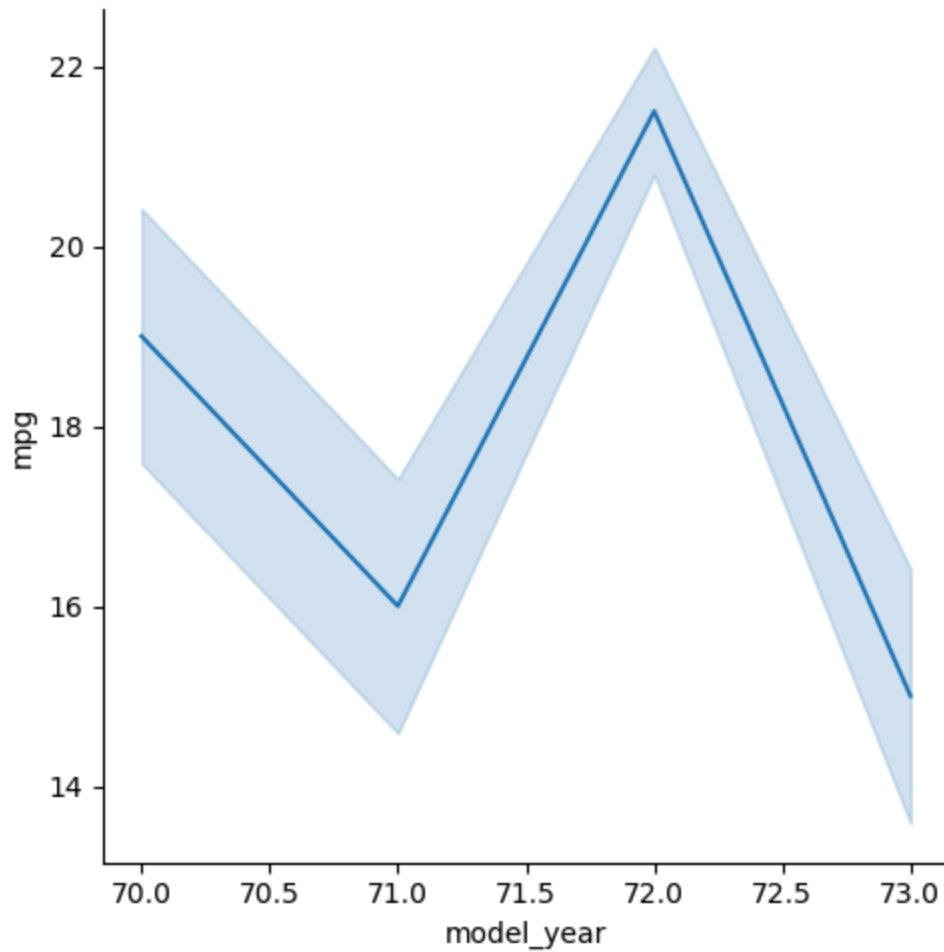



```
In [9]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

mpg = pd.DataFrame({
    "model_year": [70, 70, 71, 71, 72, 72, 73, 73],
    "mpg": [18, 20, 15, 17, 22, 21, 14, 16]
})

sns.relplot(x="model_year", y="mpg",
            data=mpg, kind="line",
            errorbar="sd")

plt.show()
```

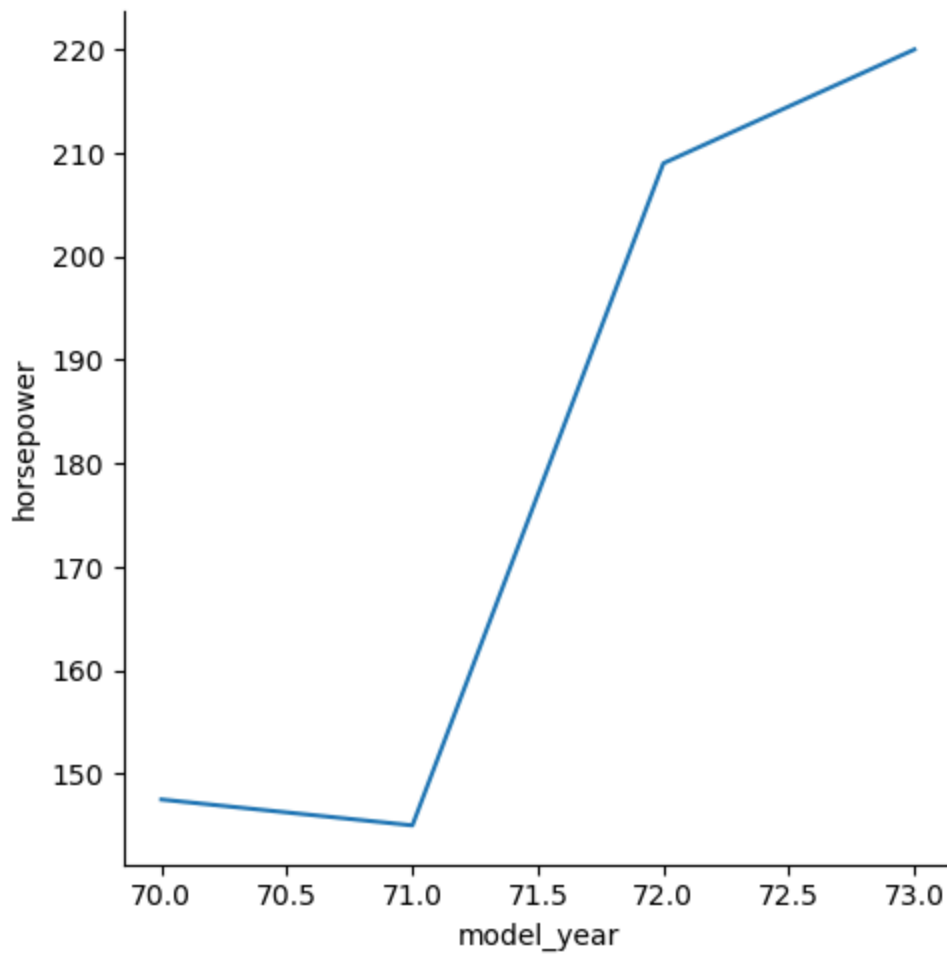


```
In [10]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

mpg = pd.DataFrame({
    "model_year": [70, 70, 71, 71, 72, 72, 73, 73],
    "horsepower": [130, 165, 150, 140, 198, 220, 215, 225]
})

sns.relplot(x="model_year", y="horsepower",
            data=mpg, kind="line",
            errorbar=None)

plt.show()
```

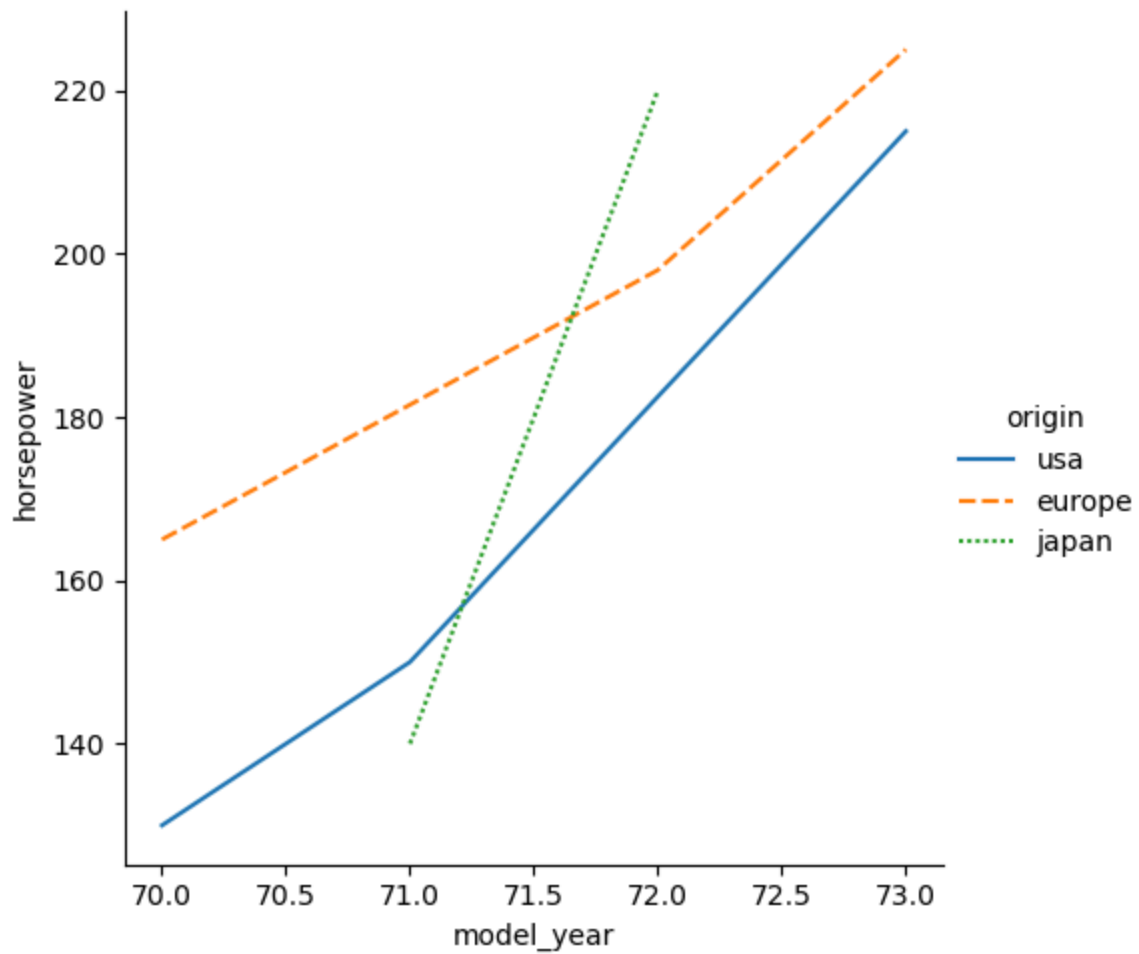


```
In [11]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

mpg = pd.DataFrame({
    "model_year": [70, 70, 71, 71, 72, 72, 73, 73],
    "horsepower": [130, 165, 150, 140, 198, 220, 215, 225],
    "origin": ["usa", "europe", "usa", "japan", "europe", "japan", "usa", "europe"]
})

sns.relplot(x="model_year", y="horsepower",
            data=mpg, kind="line",
            errorbar=None, style="origin",
            hue="origin")

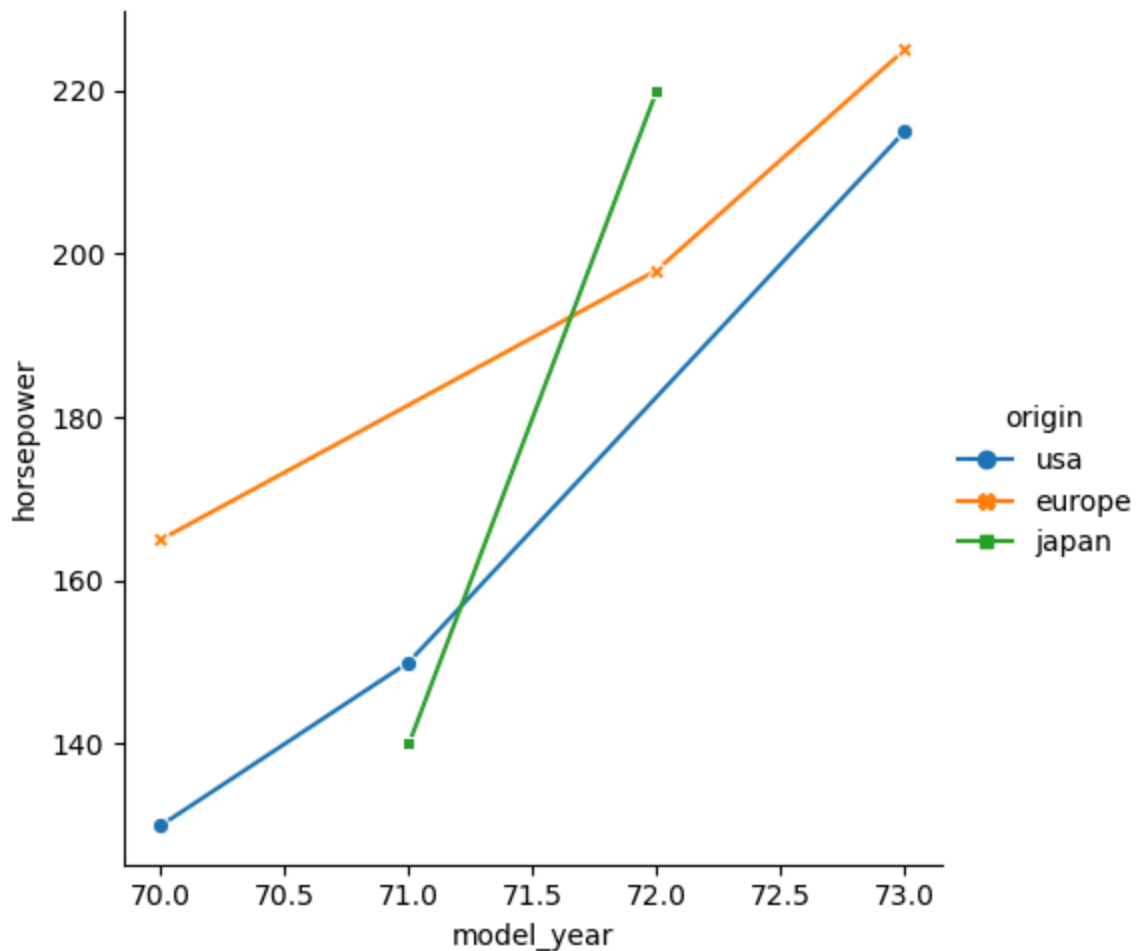
plt.show()
```



```
In [12]: import matplotlib.pyplot as plt
import seaborn as sns

sns.relplot(x="model_year", y="horsepower",
            data=mpg, kind="line",
            errorbar=None, style="origin",
            hue="origin", markers=True,
            dashes=False)

plt.show()
```



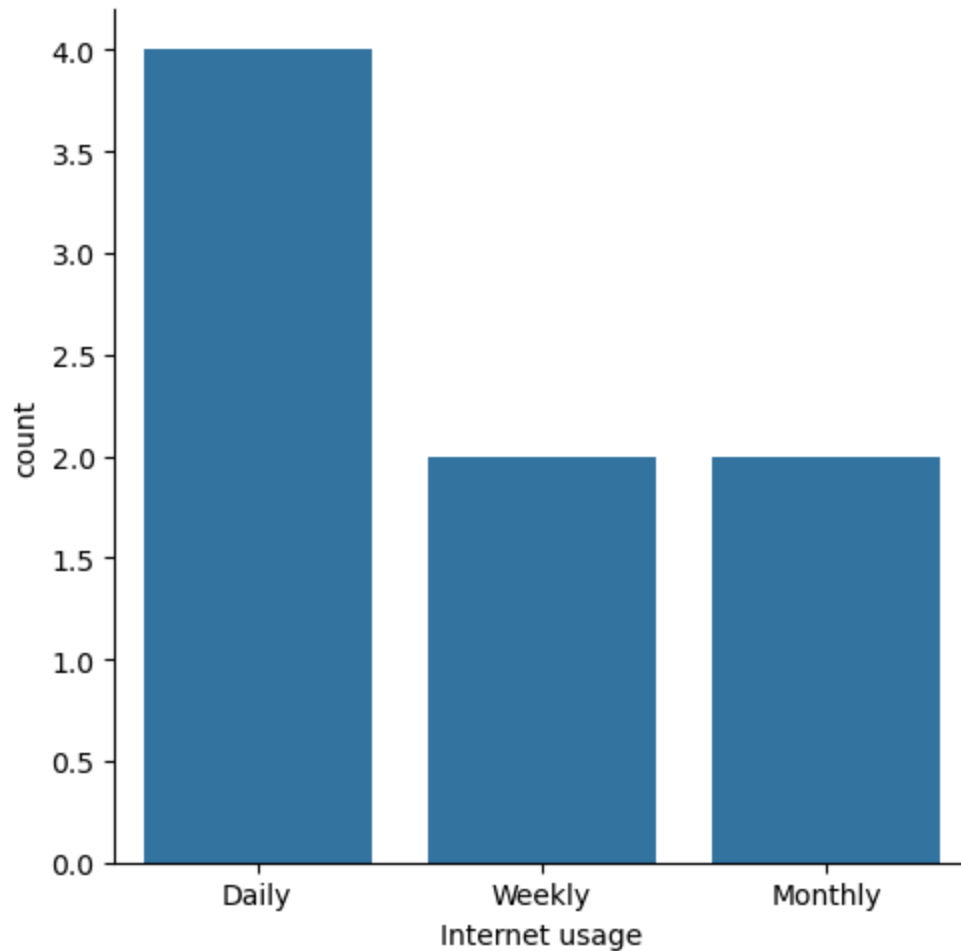
COUNT PLOTS AND BAR PLOTS:

```
In [13]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

survey_data = pd.DataFrame({
    "Internet usage": ["Daily", "Weekly", "Daily", "Monthly", "Daily", "Weekly", "M
})

sns.catplot(x="Internet usage", data=survey_data,
            kind="count")

plt.show()
```

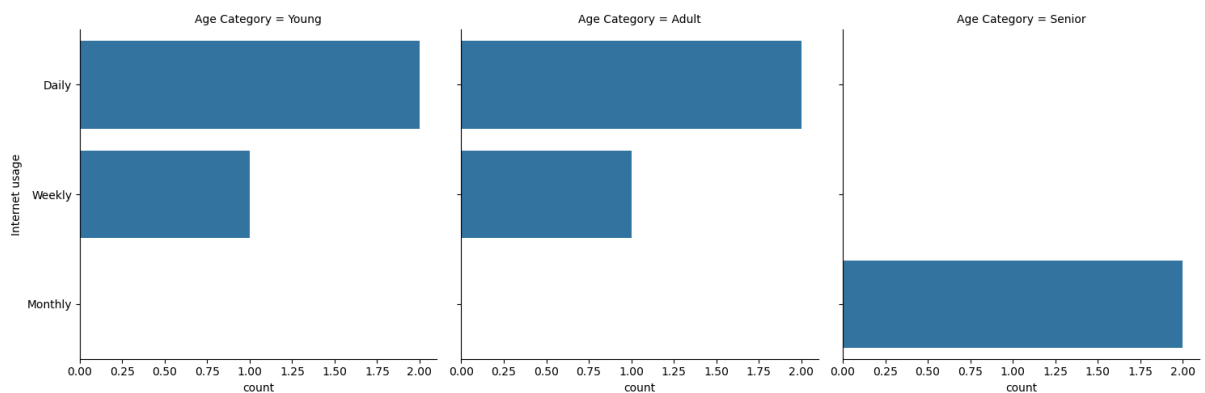


```
In [14]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

survey_data = pd.DataFrame({
    "Internet usage": ["Daily", "Weekly", "Daily", "Monthly", "Daily", "Weekly", "M
    "Age Category": ["Young", "Adult", "Young", "Senior", "Adult", "Young", "Senior
})

sns.catplot(y="Internet usage", data=survey_data,
            kind="count", col="Age Category")

plt.show()
```

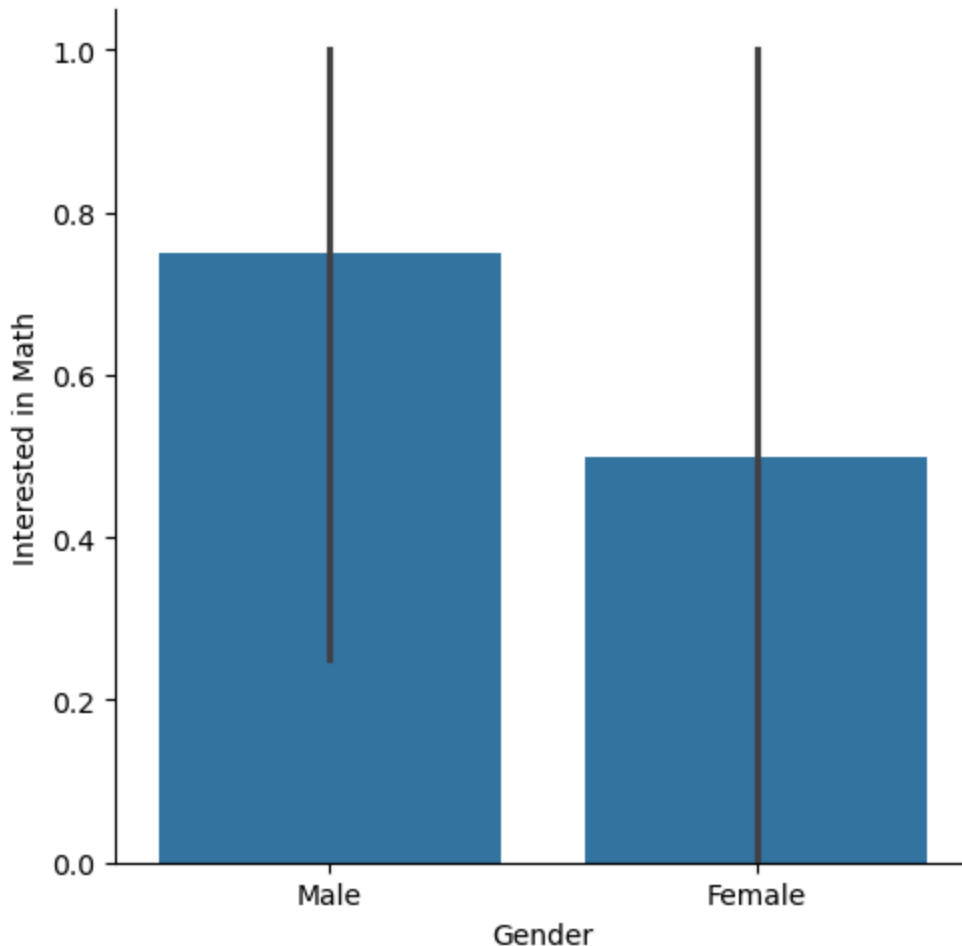


```
In [15]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

survey_data = pd.DataFrame({
    "Gender": ["Male", "Female", "Female", "Male", "Male", "Female", "Male", "Female", "Male"],
    "Interested in Math": [1, 0, 1, 1, 0, 1, 1, 0, 1]
})

sns.catplot(x="Gender", y="Interested in Math",
            data=survey_data, kind="bar")

plt.show()
```

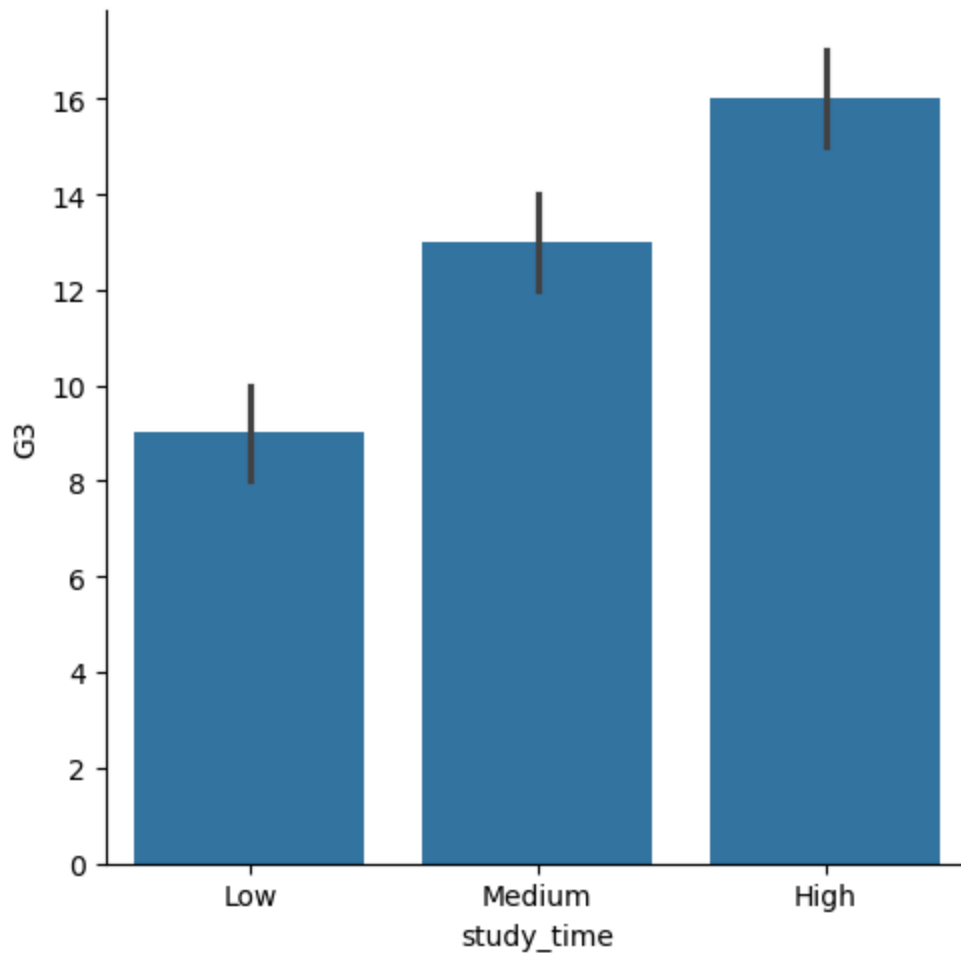


```
In [16]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

student_data = pd.DataFrame({
    "study_time": ["Low", "Medium", "High", "Low", "Medium", "High", "Low", "High"],
    "G3": [10, 12, 15, 9, 14, 16, 8, 17]
})

sns.catplot(x="study_time", y="G3",
            data=student_data,
            kind="bar")
```

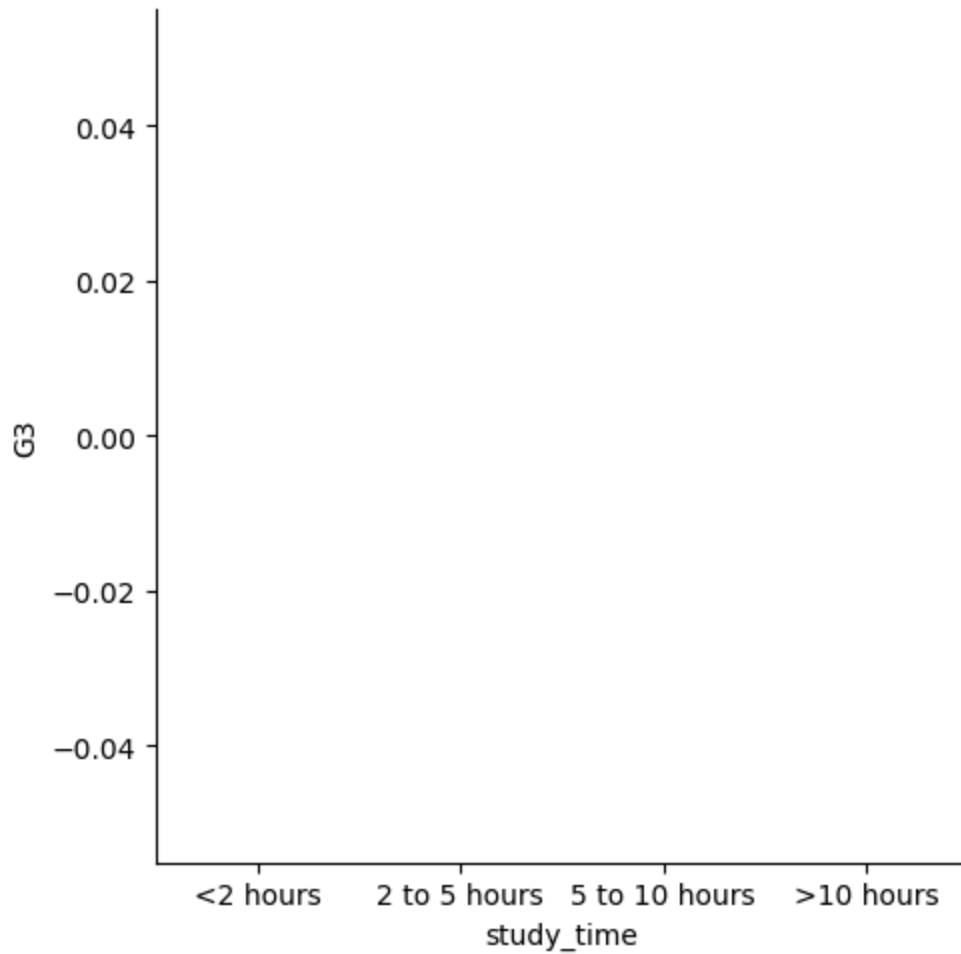
```
plt.show()
```



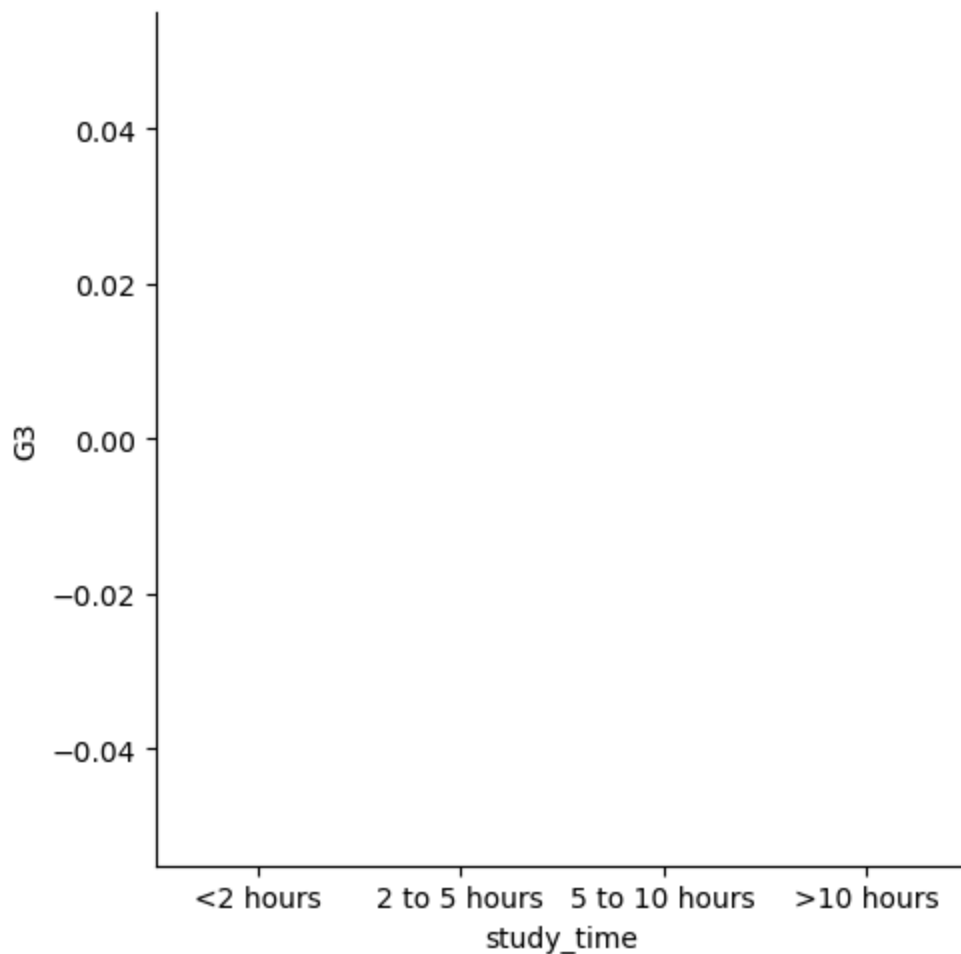
```
In [17]: student_data = pd.DataFrame({
    "study_time": ["Low", "Medium", "High", "Low", "Medium", "High", "Low", "High"]
    "G3": [10, 12, 15, 9, 14, 16, 8, 17]
})
category_order = ["<2 hours",
                  "2 to 5 hours",
                  "5 to 10 hours",
                  ">10 hours"]

# Rearrange the categories
sns.catplot(x="study_time", y="G3",
            data=student_data,
            kind="bar",
            order=category_order)

# Show plot
plt.show()
```

```
In [18]: category_order = ["<2 hours",  
                           "2 to 5 hours",  
                           "5 to 10 hours",  
                           ">10 hours"]  
  
# Turn off the confidence intervals  
sns.catplot(x="study_time", y="G3",  
            data=student_data,  
            kind="bar",  
            order=category_order,  
            errorbar=None)  
  
# Show plot  
plt.show()
```



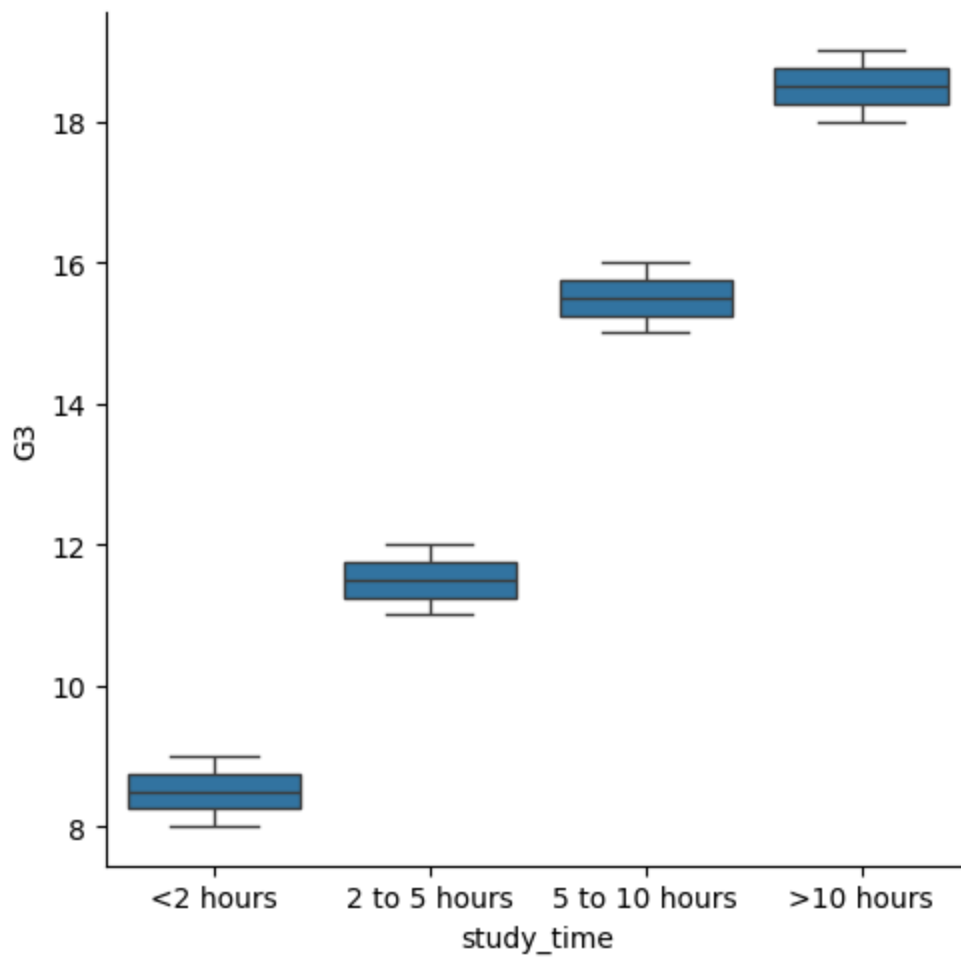
```
In [19]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

student_data = pd.DataFrame({
    "study_time": [
        "<2 hours", "2 to 5 hours", "5 to 10 hours", ">10 hours",
        "<2 hours", "2 to 5 hours", "5 to 10 hours", ">10 hours"
    ],
    "G3": [8, 12, 15, 18, 9, 11, 16, 19]
})

study_time_order = ["<2 hours", "2 to 5 hours", "5 to 10 hours", ">10 hours"]

sns.catplot(x="study_time", y="G3",
            data=student_data,
            kind="box",
            order=study_time_order)

plt.show()
```

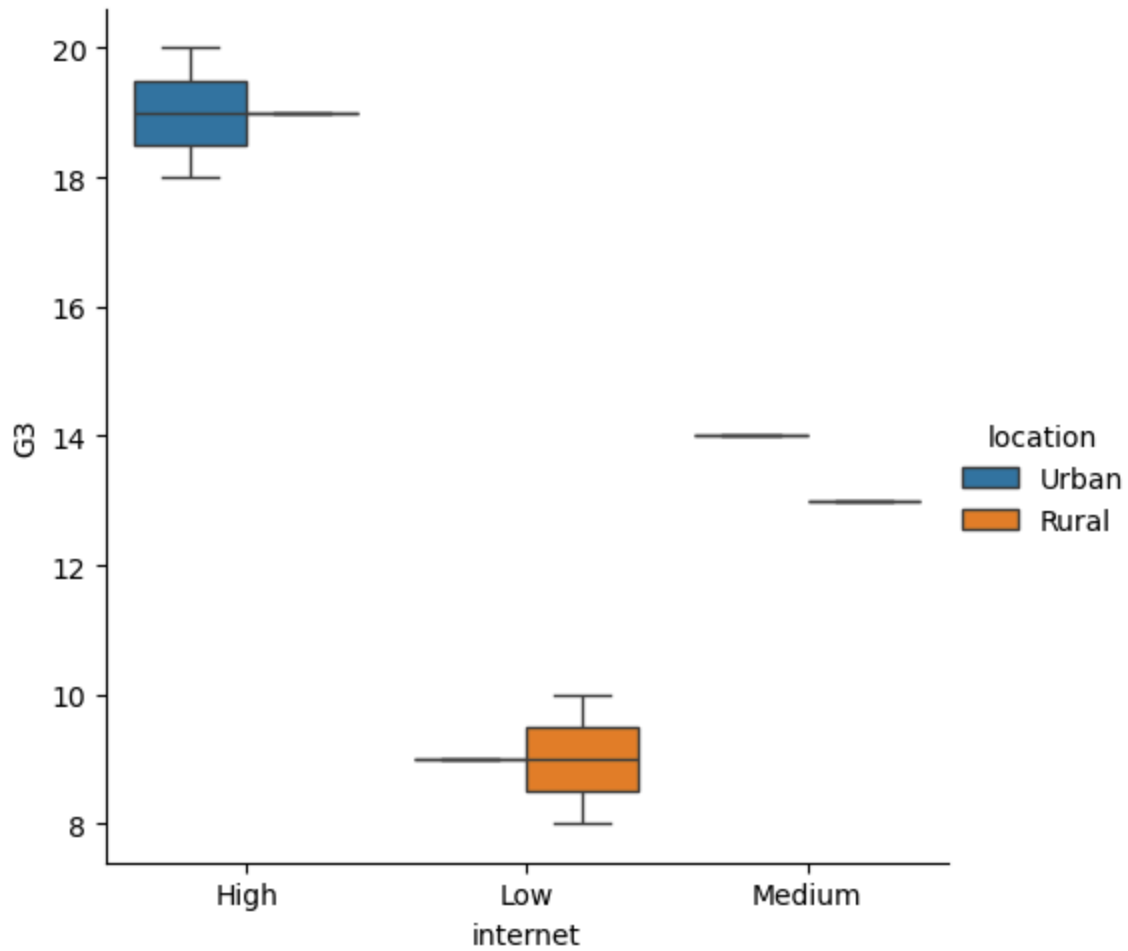


```
In [20]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

student_data = pd.DataFrame({
    "internet": ["High", "Low", "Medium", "High", "Low", "Medium", "High", "Low"],
    "G3": [18, 10, 14, 19, 9, 13, 20, 8],
    "location": ["Urban", "Rural", "Urban", "Rural", "Urban", "Rural", "Urban", "Ru
    })

sns.catplot(x="internet", y="G3",
            data=student_data,
            kind="box",
            hue="location",
            showfliers=False)

plt.show()
```

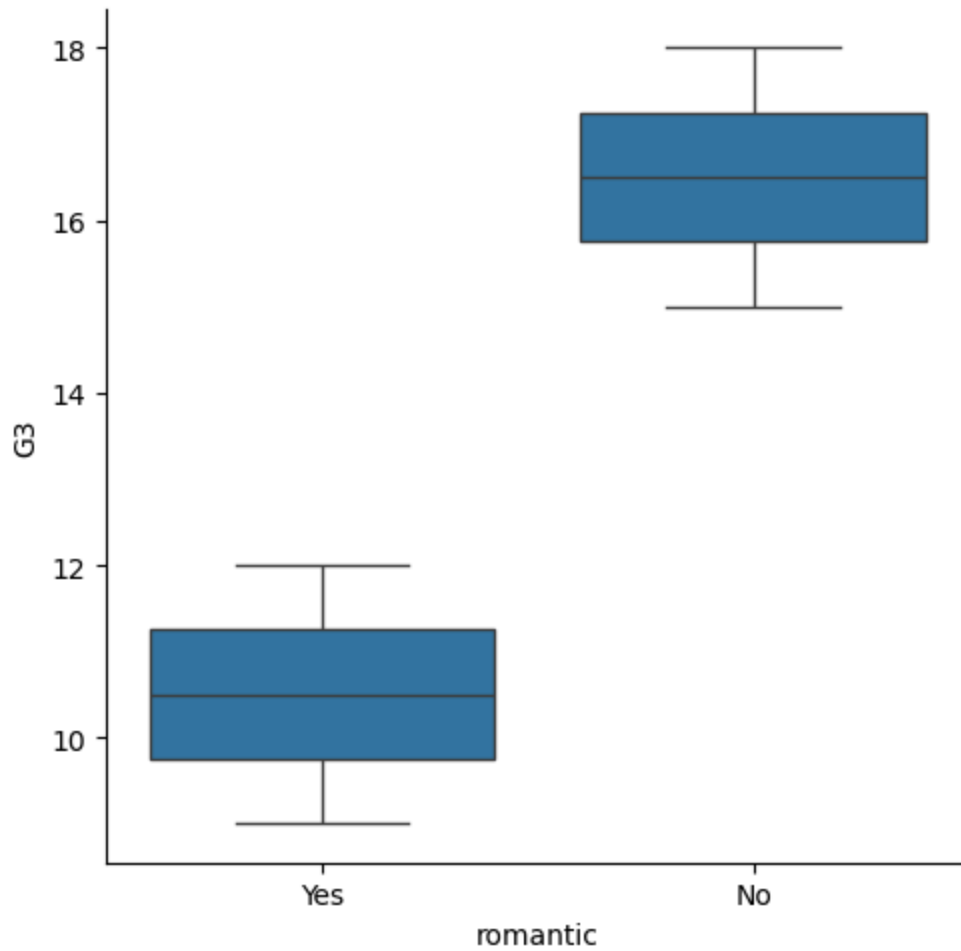


```
In [21]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

student_data = pd.DataFrame({
    "romantic": ["Yes", "No", "Yes", "No", "Yes", "No", "Yes", "No"],
    "G3": [12, 15, 10, 18, 11, 16, 9, 17]
})

sns.catplot(x="romantic", y="G3",
            data=student_data,
            kind="box",
            whis=0.5)

plt.show()
```

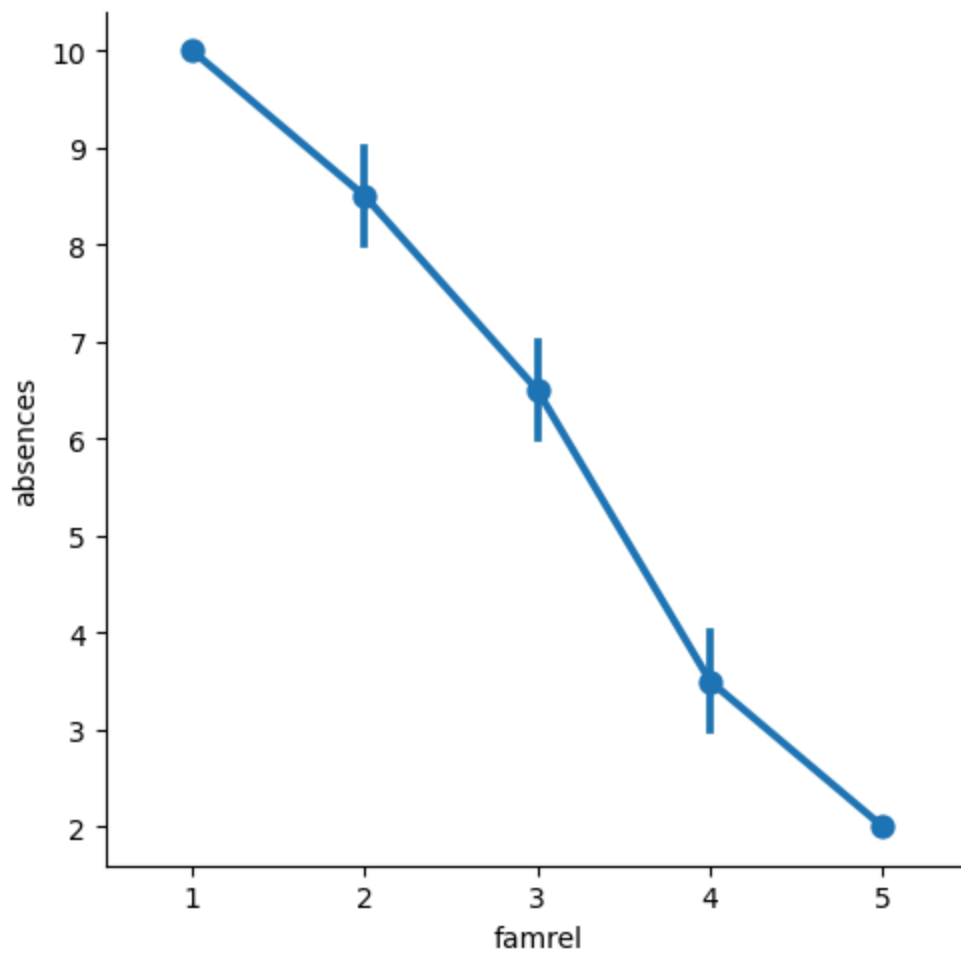


```
In [22]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

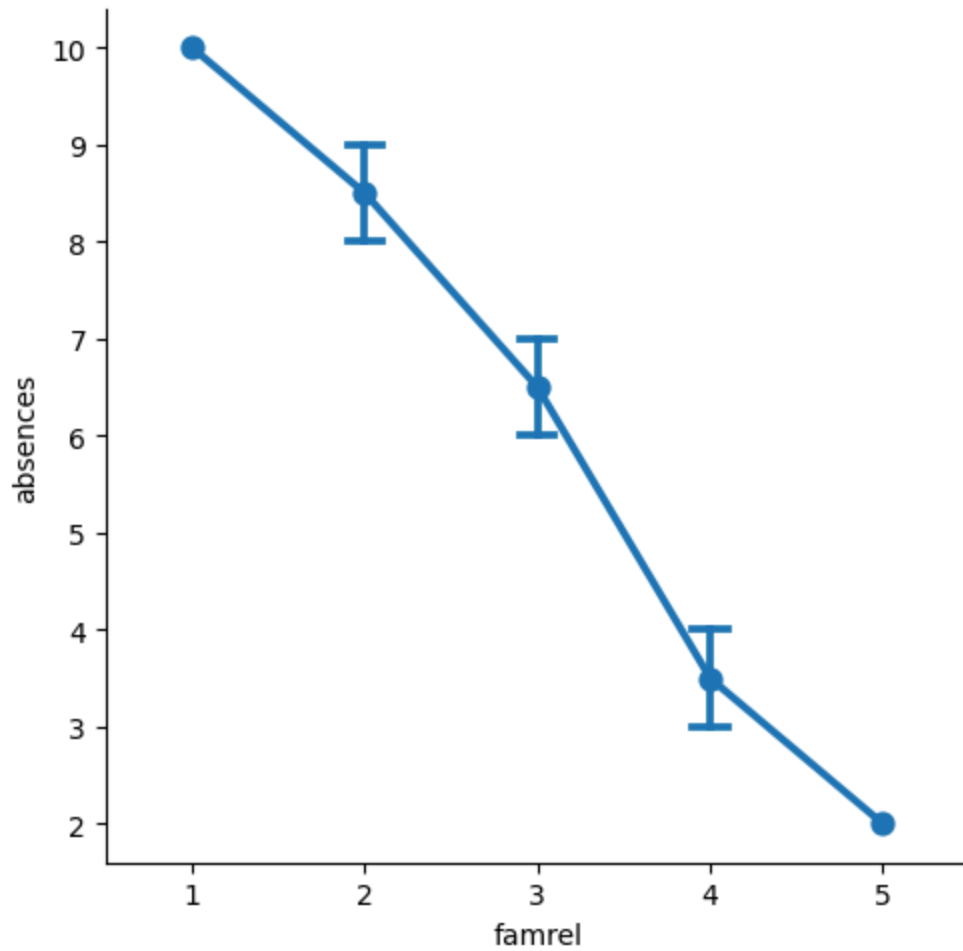
student_data = pd.DataFrame({
    "famrel": [1, 2, 3, 4, 5, 3, 2, 4],
    "absences": [10, 8, 6, 4, 2, 7, 9, 3]
})

sns.catplot(x="famrel", y="absences",
            data=student_data,
            kind="point")

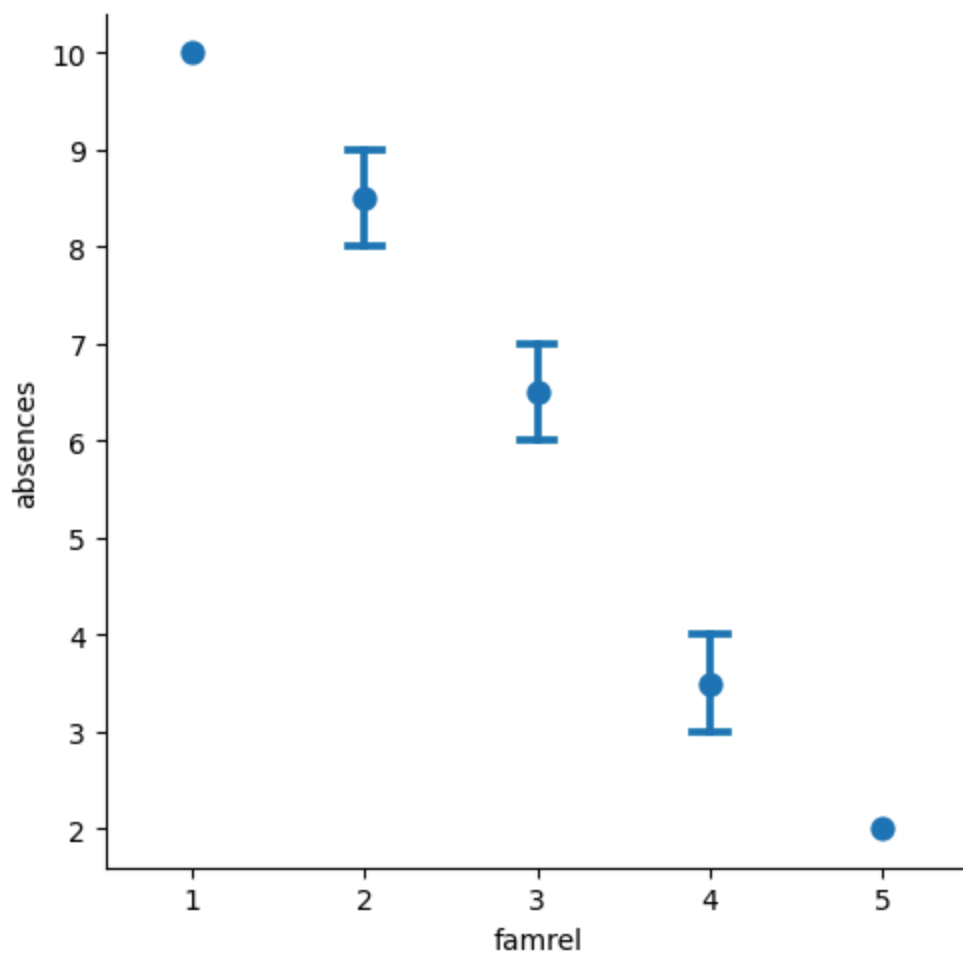
plt.show()
```



```
In [23]: sns.catplot(x="famrel", y="absences",  
                    data=student_data,  
                    kind="point",  
                    capsize=0.2)  
plt.show()
```



```
In [24]: sns.catplot(x="famrel", y="absences",  
                    data=student_data,  
                    kind="point",  
                    capsize=0.2,  
                    linestyle="none")  
plt.show()
```

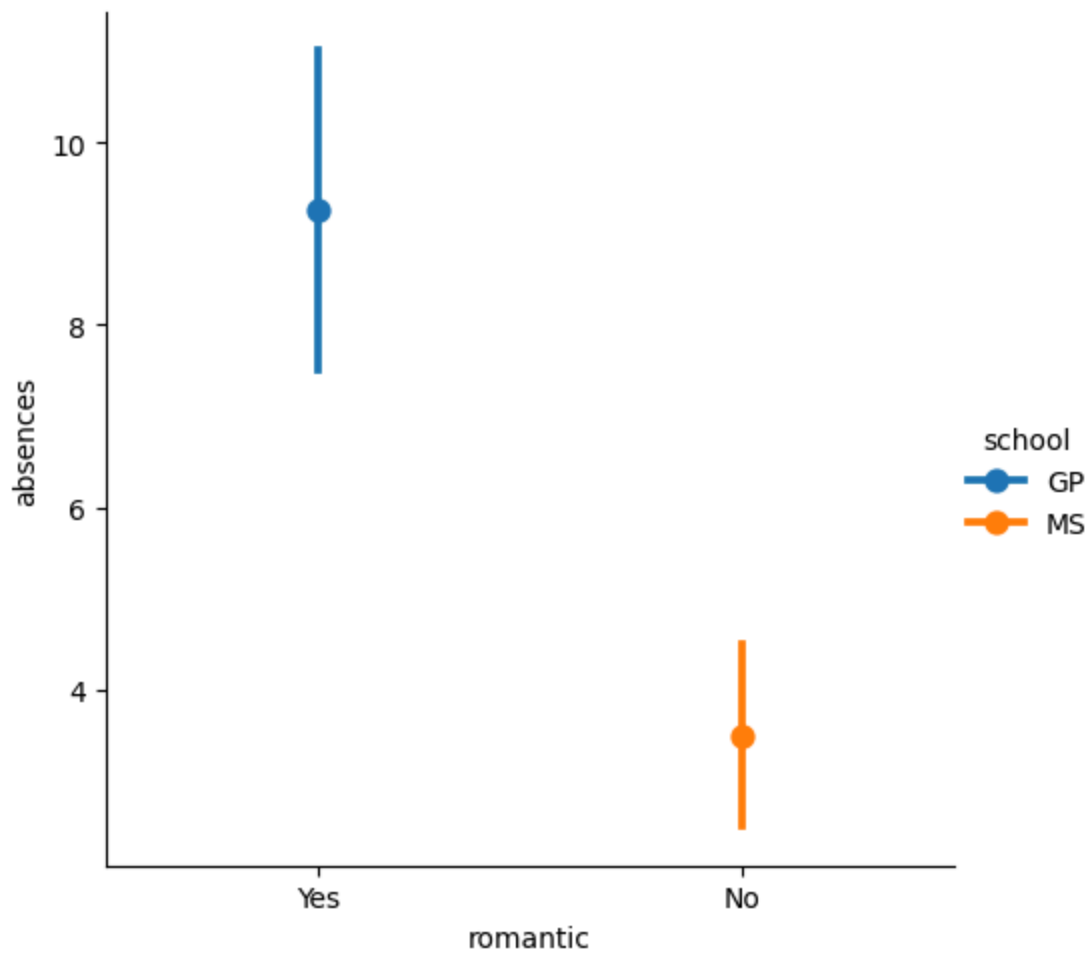


```
In [25]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

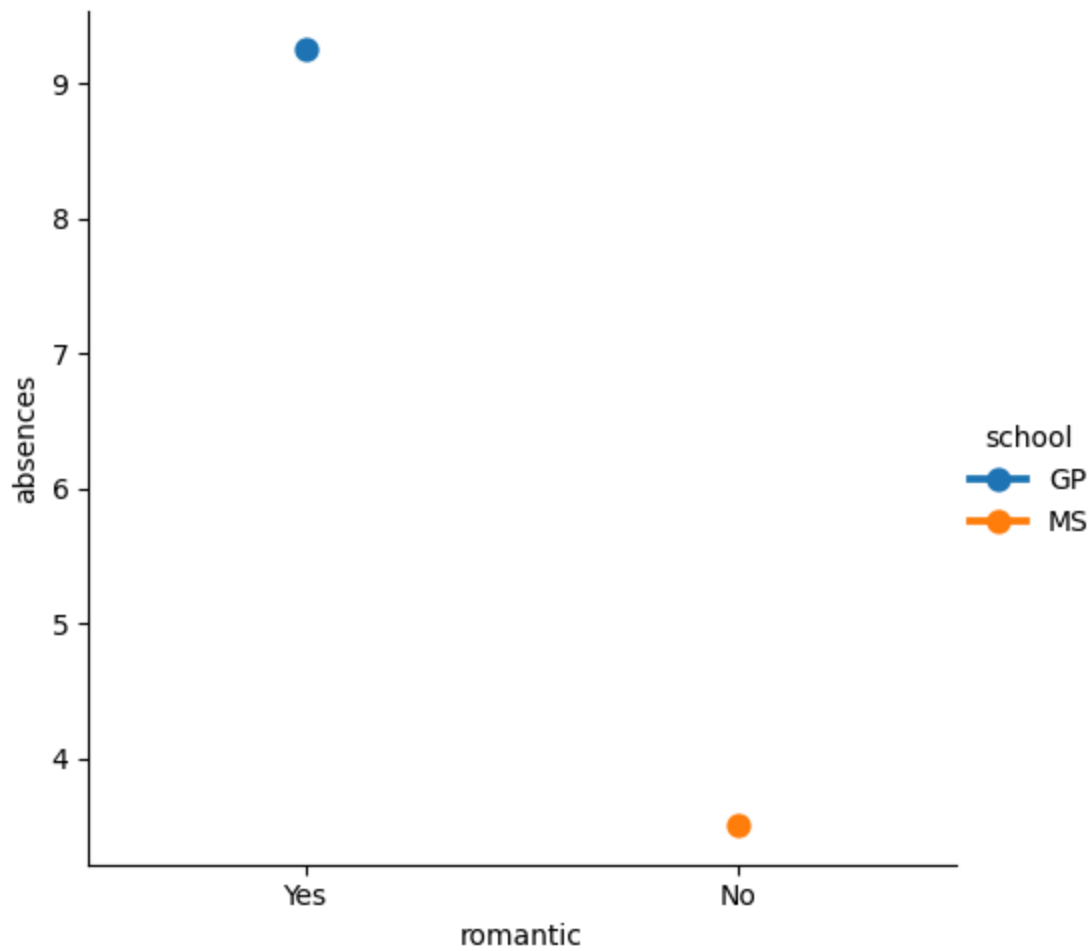
student_data = pd.DataFrame({
    "romantic": ["Yes", "No", "Yes", "No", "Yes", "No", "Yes", "No"],
    "absences": [10, 4, 8, 3, 12, 5, 7, 2],
    "school": ["GP", "MS", "GP", "MS", "GP", "MS", "GP", "MS"]
})

sns.catplot(x="romantic", y="absences",
            data=student_data,
            kind="point",
            hue="school")

plt.show()
```

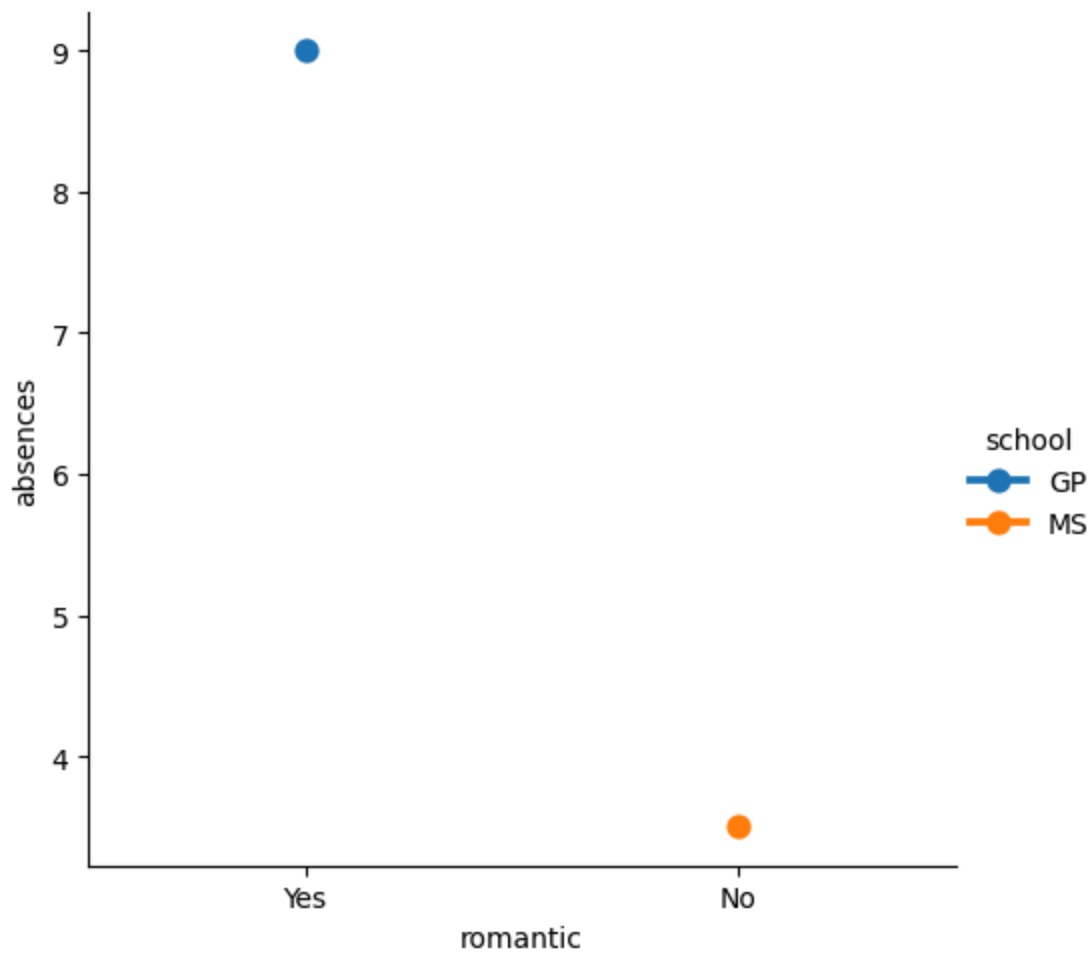
```
In [26]: sns.catplot(x="romantic", y="absences",  
                    data=student_data,  
                    kind="point",  
                    hue="school",  
                    errorbar=None)  
plt.show()
```



```
In [27]: from numpy import median

sns.catplot(x="romantic", y="absences",
            data=student_data,
            kind="point",
            hue="school",
            errorbar=None,
            estimator=median)

plt.show()
```



CHANGING PLOT STYLE AND COLOR:

```
In [28]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

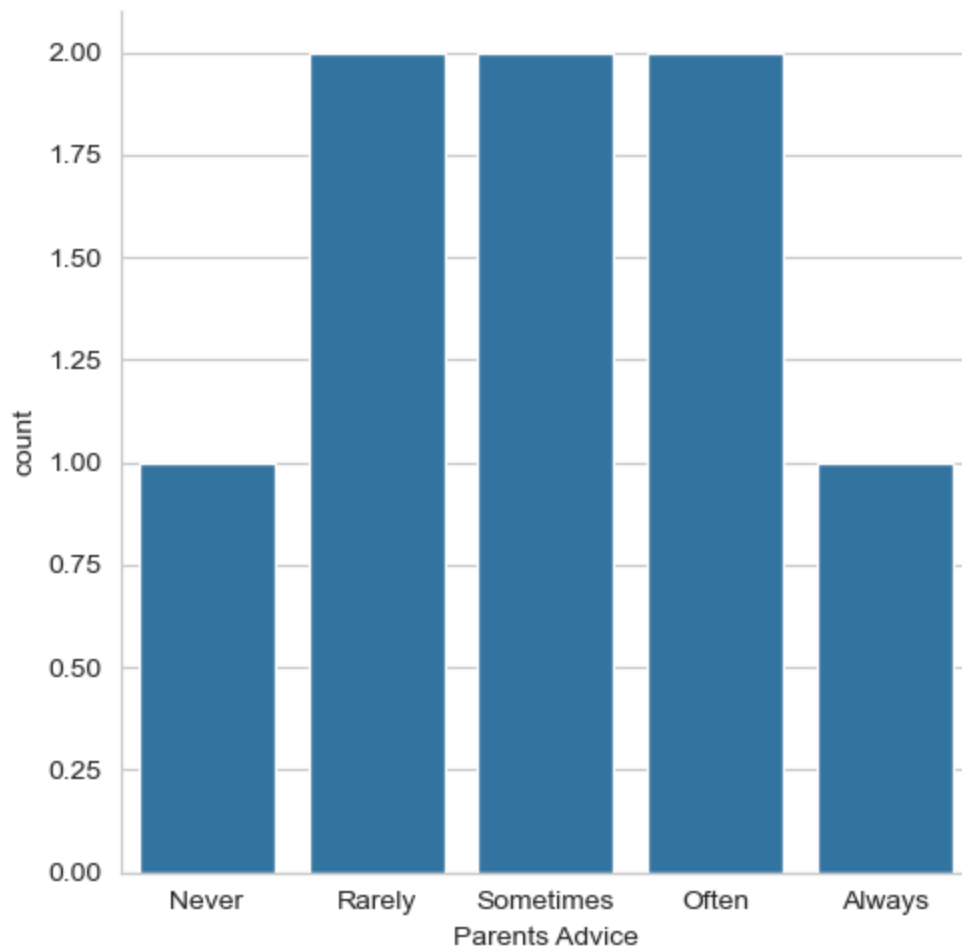
sns.set_style("whitegrid")

survey_data = pd.DataFrame({
    "Parents Advice": [
        "Never", "Rarely", "Sometimes", "Often",
        "Always", "Sometimes", "Rarely", "Often"
    ]
})

category_order = ["Never", "Rarely", "Sometimes", "Often", "Always"]

sns.catplot(x="Parents Advice",
            data=survey_data,
            kind="count",
            order=category_order)

plt.show()
```

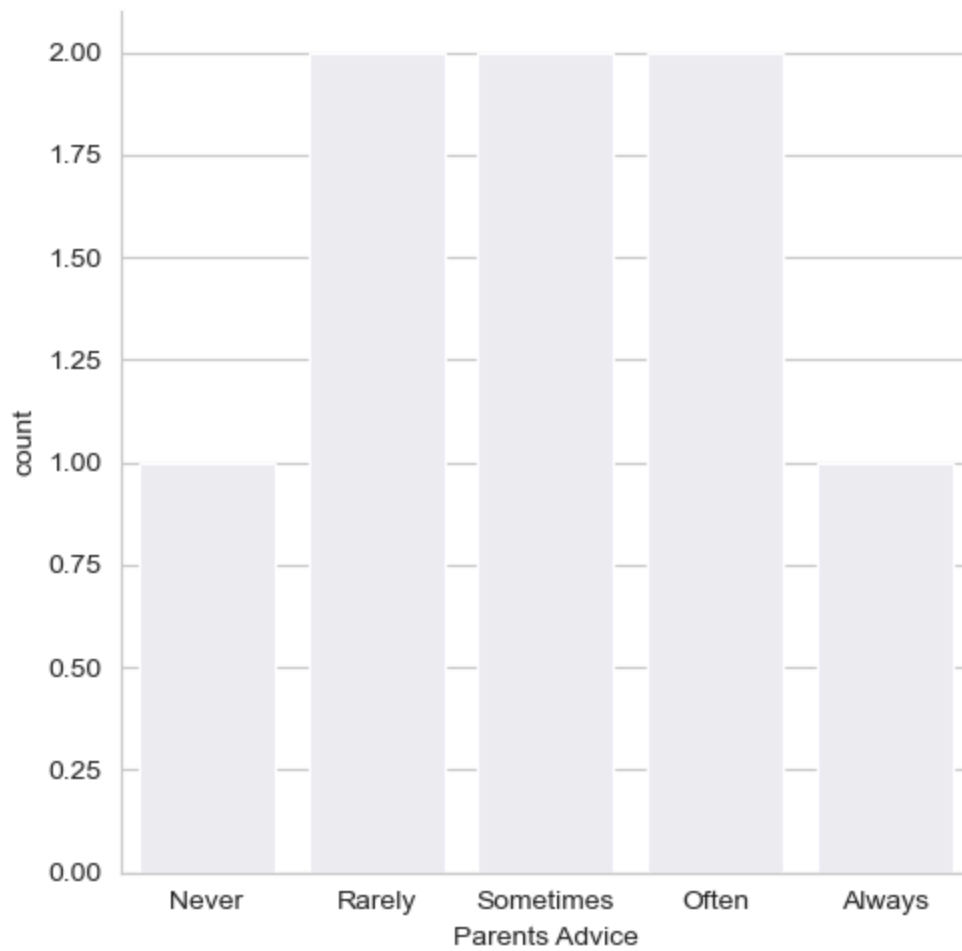


```
In [29]: sns.set_style("whitegrid")
sns.set_palette("Purples")

category_order = ["Never", "Rarely", "Sometimes",
                  "Often", "Always"]

sns.catplot(x="Parents Advice",
            data=survey_data,
            kind="count",
            order=category_order)

plt.show()
```

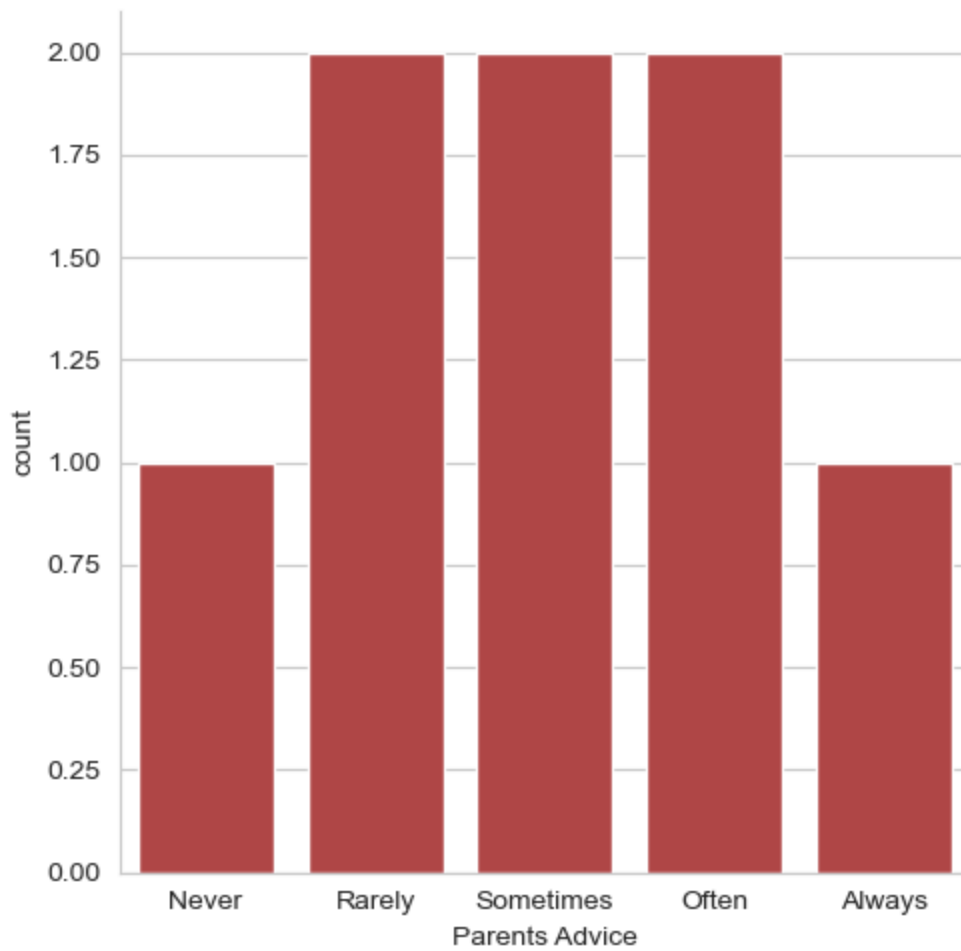


```
In [30]: sns.set_style("whitegrid")
sns.set_palette("RdBu")

category_order = ["Never", "Rarely", "Sometimes",
                  "Often", "Always"]

sns.catplot(x="Parents Advice",
            data=survey_data,
            kind="count",
            order=category_order)

plt.show()
```



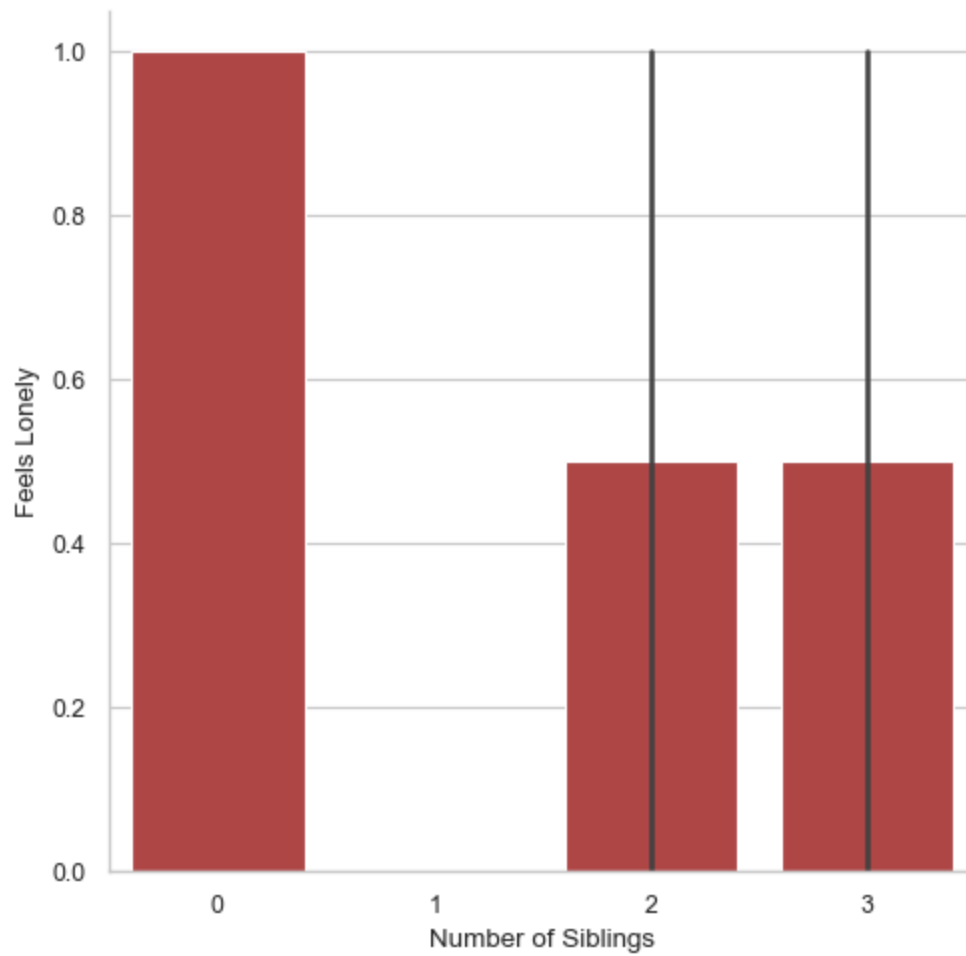
```
In [31]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

sns.set_context("paper")

survey_data = pd.DataFrame({
    "Number of Siblings": [0, 1, 2, 3, 1, 2, 0, 3],
    "Feels Lonely": [1, 0, 1, 1, 0, 0, 1, 0]
})

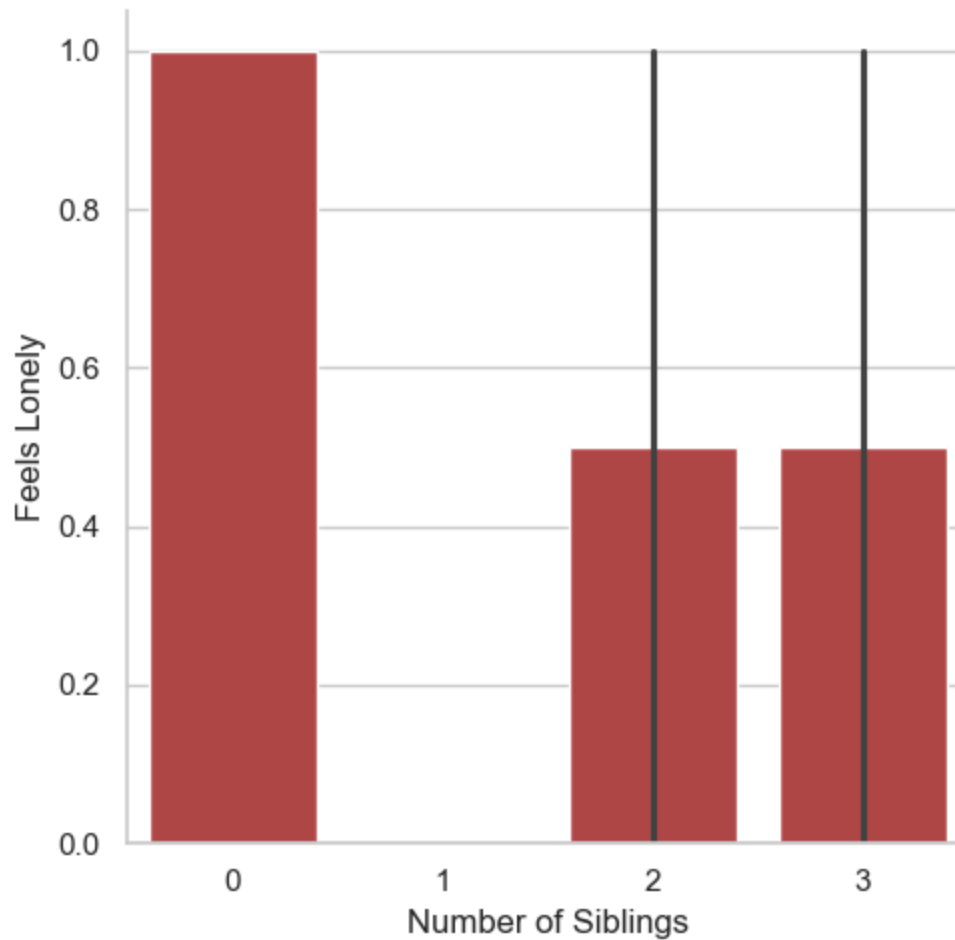
sns.catplot(x="Number of Siblings", y="Feels Lonely",
            data=survey_data, kind="bar")

plt.show()
```



```
In [32]: sns.set_context("notebook")

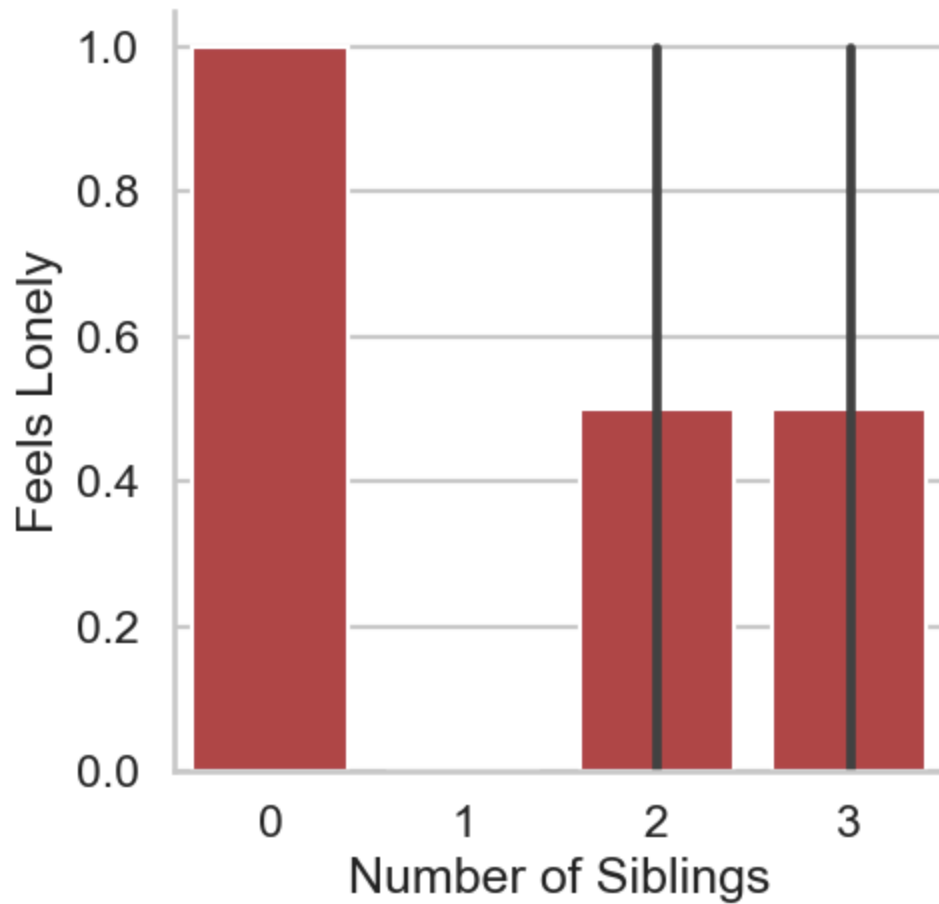
sns.catplot(x="Number of Siblings", y="Feels Lonely",
            data=survey_data, kind="bar")
plt.show()
```



```
In [33]: sns.set_context("talk")

sns.catplot(x="Number of Siblings", y="Feels Lonely",
            data=survey_data, kind="bar")

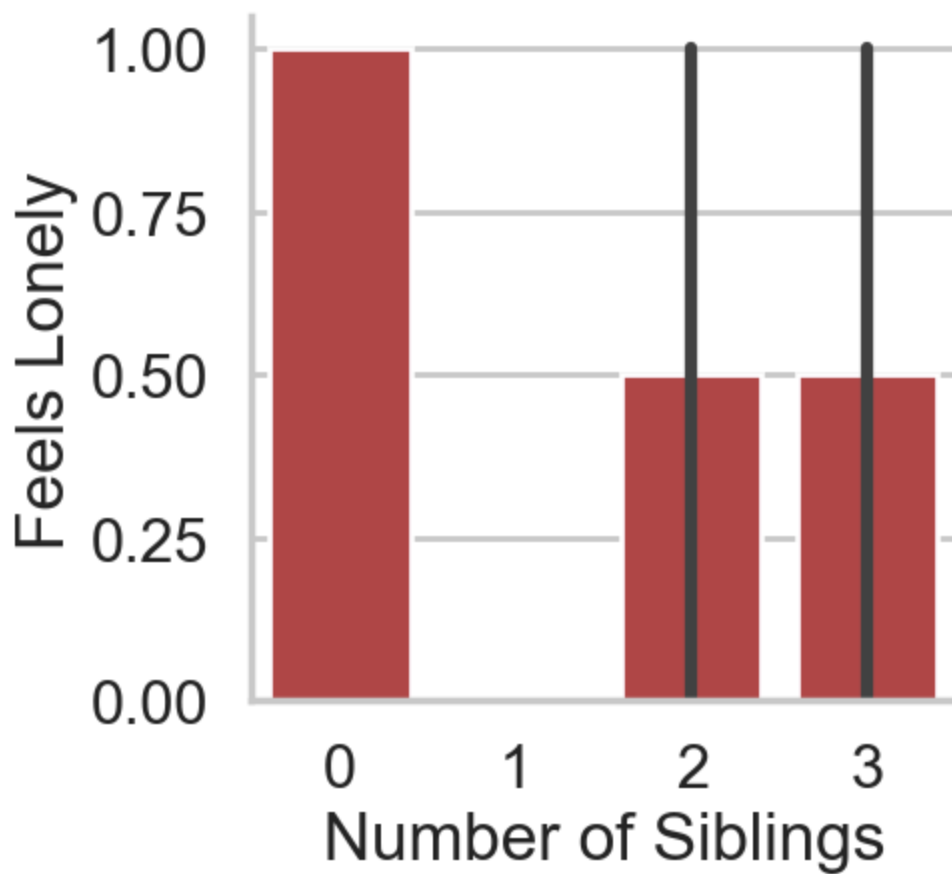
plt.show()
```

```
In [34]: sns.set_context("poster")

sns.catplot(x="Number of Siblings", y="Feels Lonely",
            data=survey_data, kind="bar")

plt.show()
```



```
In [35]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

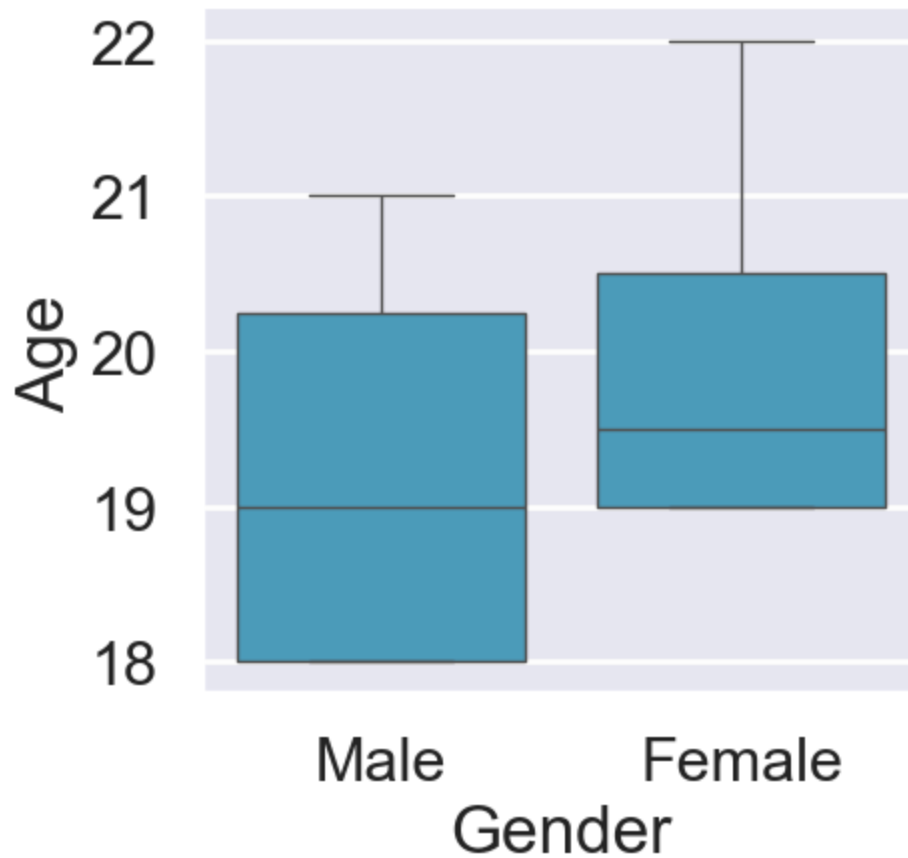
sns.set_style("darkgrid")

# custom palette
sns.set_palette(["#39A7D0", "#36ADA4"])

# dataset
survey_data = pd.DataFrame({
    "Gender": ["Male", "Female", "Female", "Male", "Male", "Female", "Male", "Female"],
    "Age": [18, 19, 20, 18, 21, 19, 20, 22]
})

# box plot
sns.catplot(x="Gender", y="Age",
            data=survey_data, kind="box")

plt.show()
```



```
In [36]: import matplotlib.pyplot as plt
import seaborn as sns
```

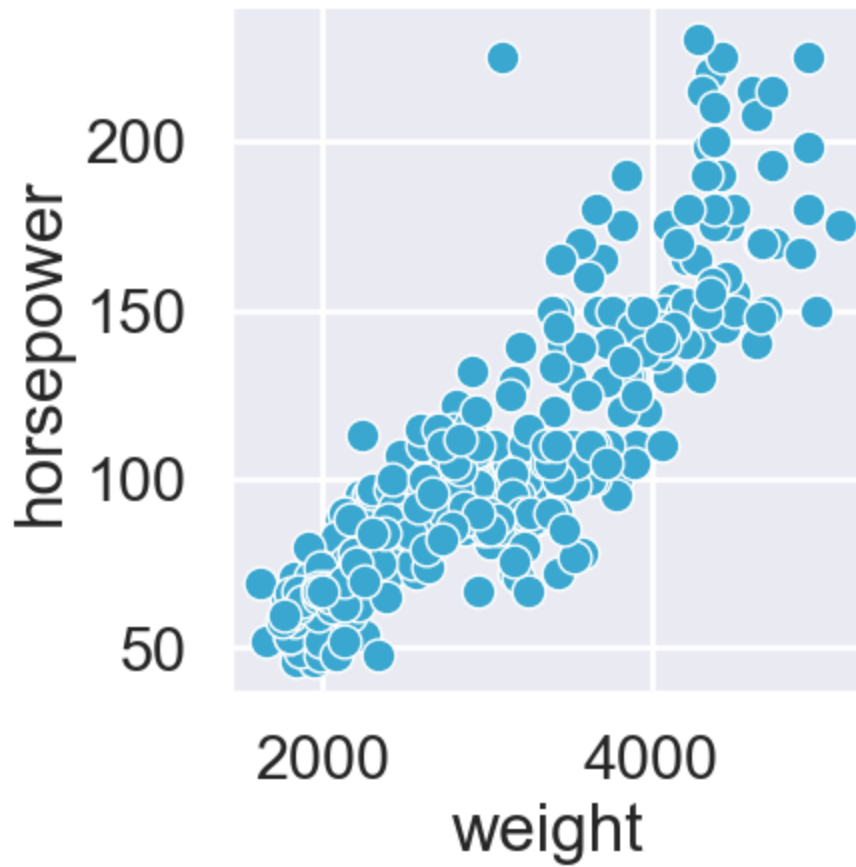
```
mpg = sns.load_dataset("mpg")
mpg = mpg.dropna()
```

```
g = sns.relplot(x="weight",
                y="horsepower",
                data=mpg,
                kind="scatter")
```

```
print(type(g))
```

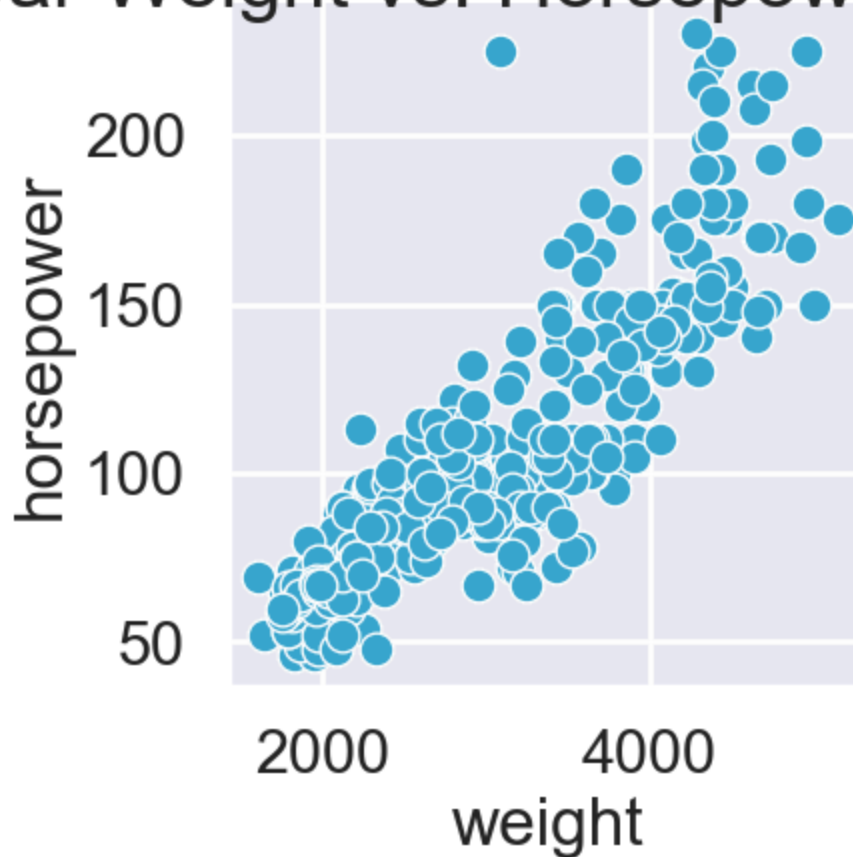
```
plt.show()
```

```
<class 'seaborn.axisgrid.FacetGrid'>
```



```
In [37]: g = sns.relplot(x="weight",  
                        y="horsepower",  
                        data=mpg,  
                        kind="scatter")  
  
g.figure.suptitle("Car Weight vs. Horsepower")  
  
plt.show()
```

Car Weight vs. Horsepower



```
In [38]: import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

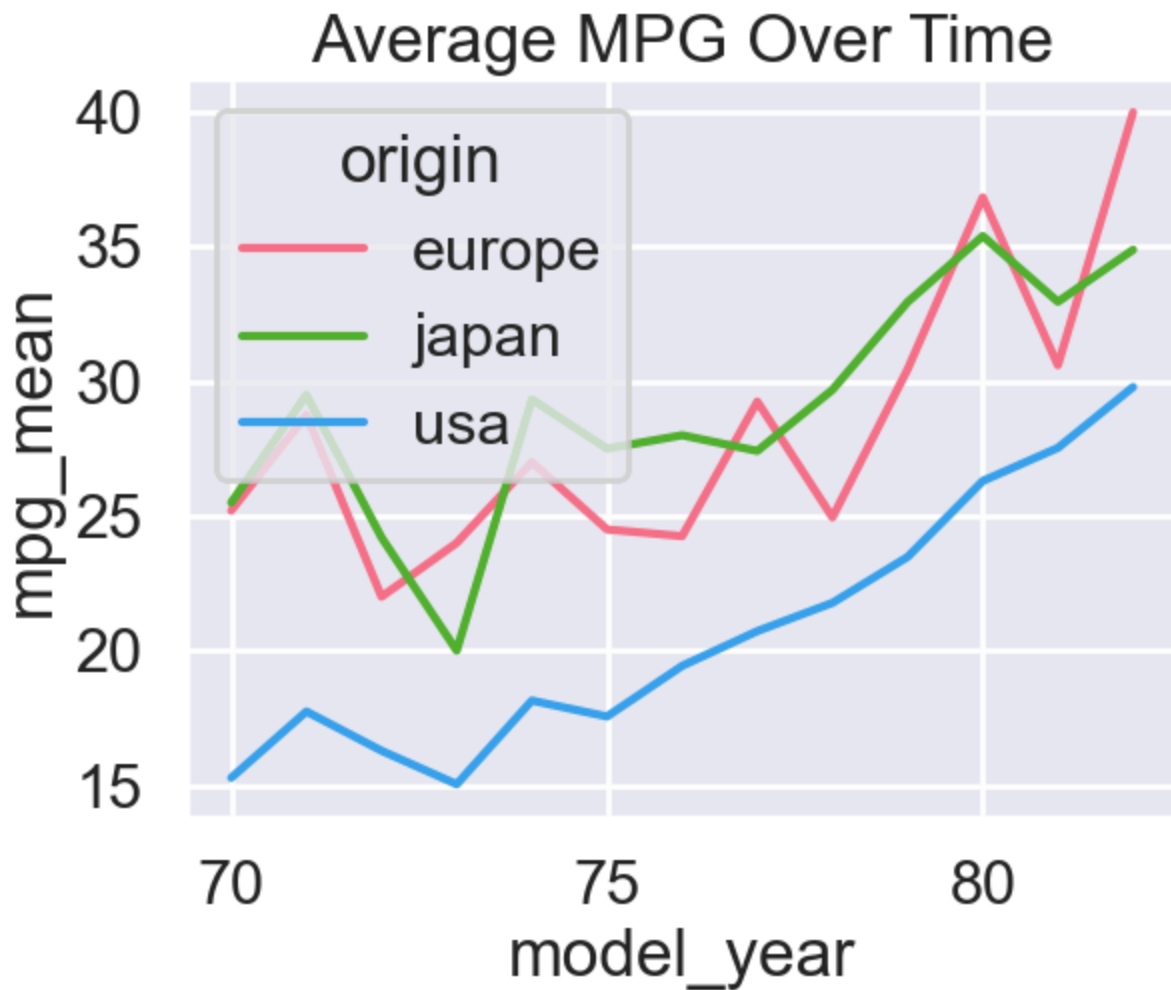
mpg = sns.load_dataset("mpg")
mpg = mpg.dropna()

mpg_mean = mpg.groupby(["model_year", "origin"])["mpg"].mean().reset_index()
mpg_mean.rename(columns={"mpg": "mpg_mean"}, inplace=True)

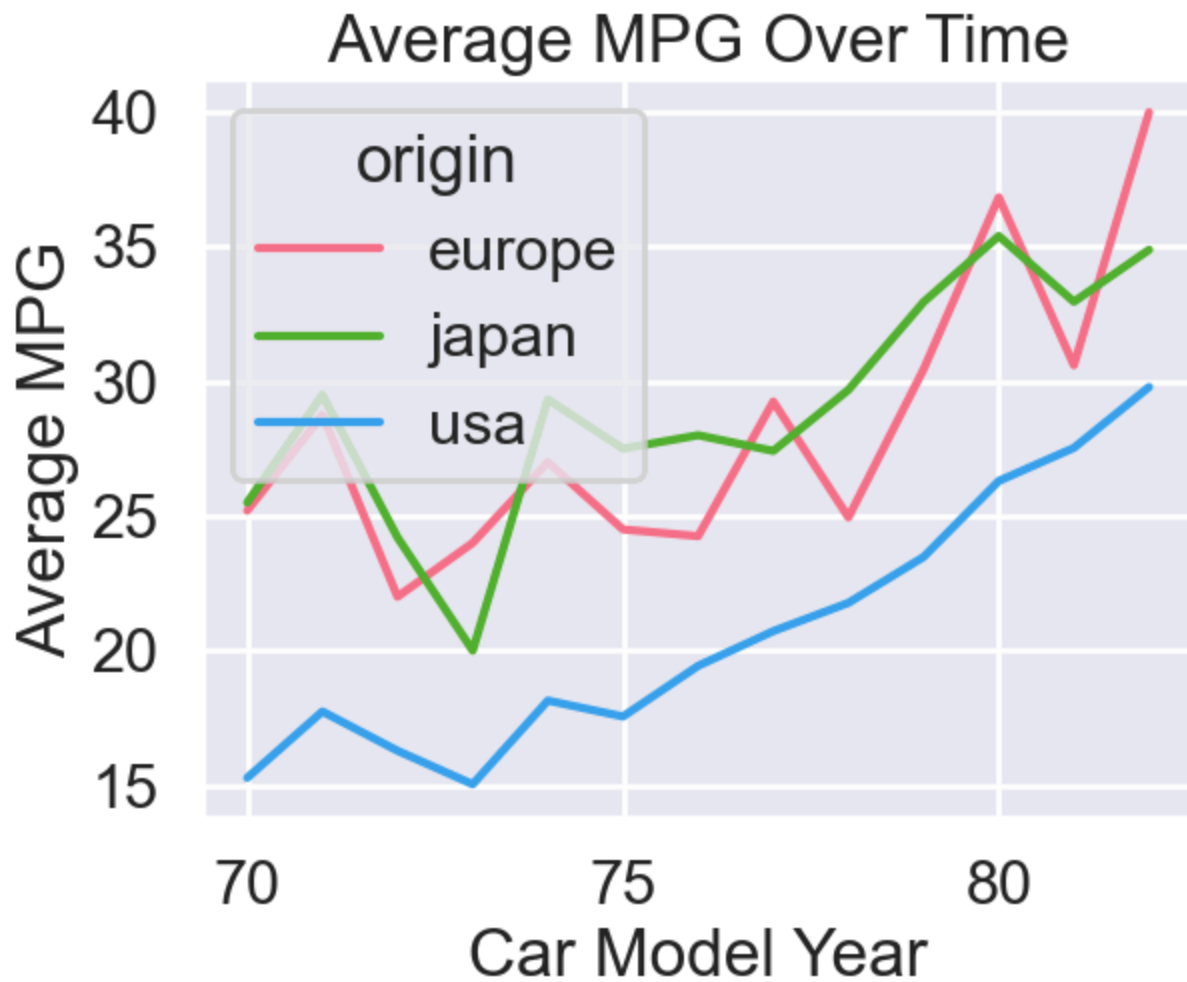
g = sns.lineplot(x="model_year", y="mpg_mean",
                  data=mpg_mean,
                  hue="origin")

g.set_title("Average MPG Over Time")

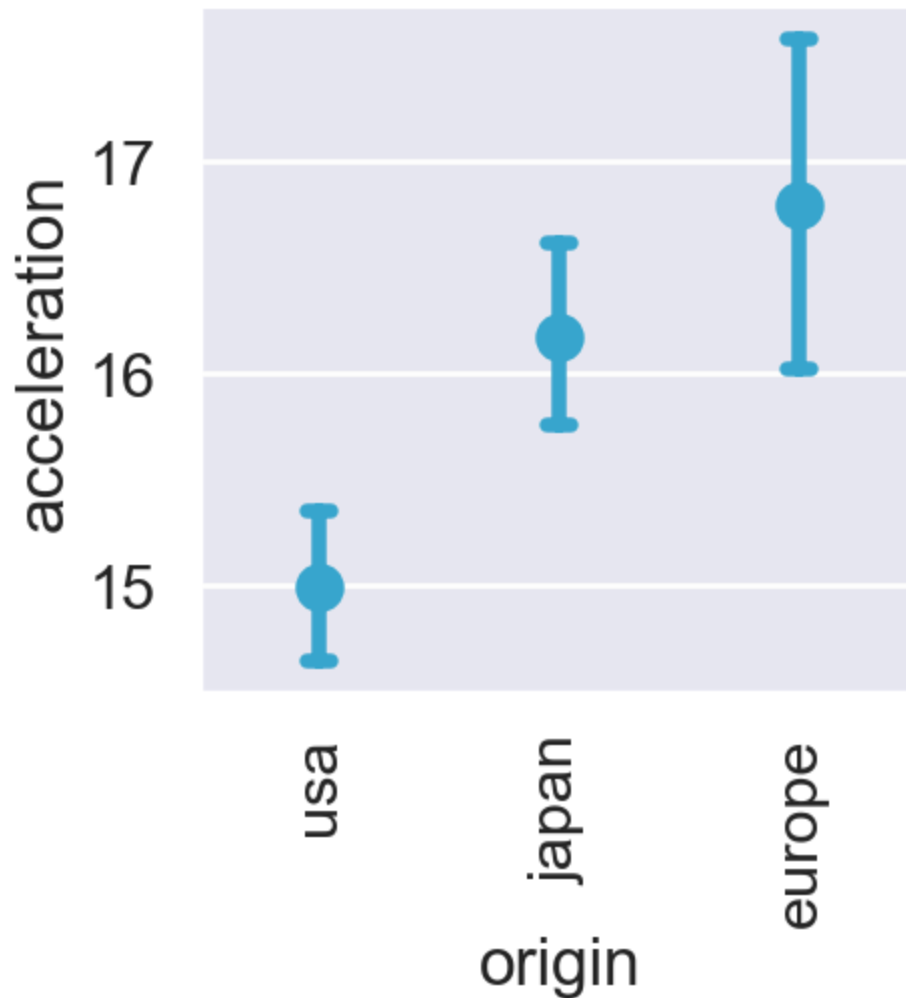
plt.show()
```



```
In [39]: g = sns.lineplot(x="model_year", y="mpg_mean",  
                        data=mpg_mean,  
                        hue="origin")  
  
g.set_title("Average MPG Over Time")  
  
g.set(xlabel="Car Model Year",  
      ylabel="Average MPG")  
  
plt.show()
```



```
In [40]: sns.catplot(x="origin",  
                    y="acceleration",  
                    data=mpg,  
                    kind="point",  
                    linestyle="none",  
                    capsize=0.1)  
  
plt.xticks(rotation=90)  
  
plt.show()
```



```
In [41]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

sns.set_palette("Blues")

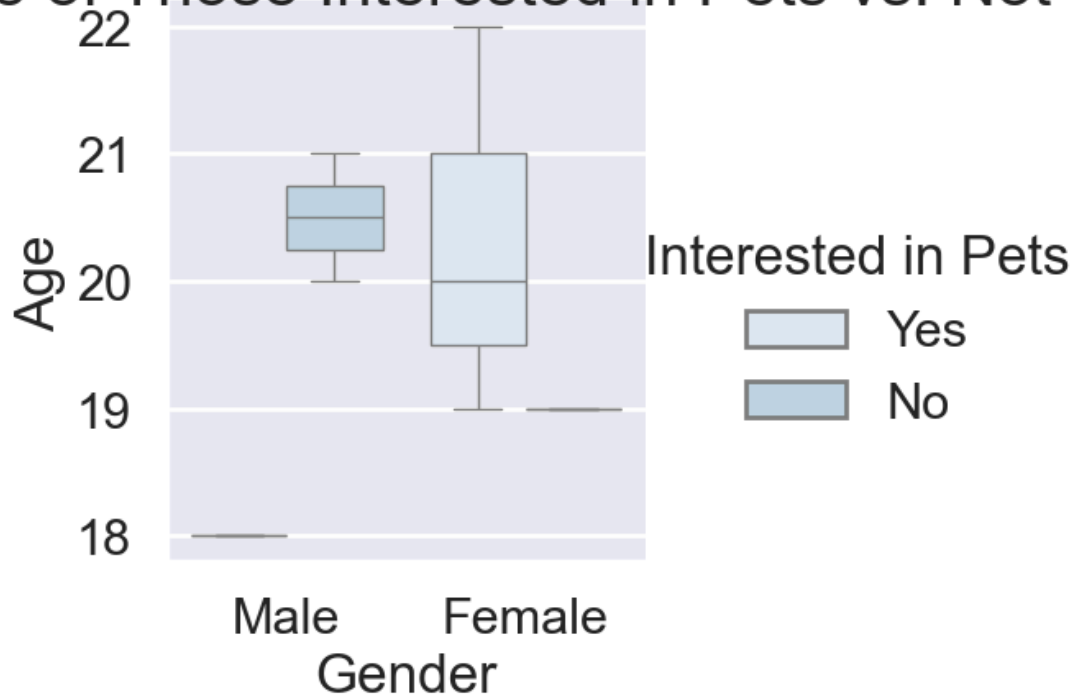
survey_data = pd.DataFrame({
    "Gender": ["Male", "Female", "Female", "Male", "Male", "Female", "Male", "Female", "Male"],
    "Age": [18, 19, 20, 18, 21, 19, 20, 22],
    "Interested in Pets": ["Yes", "No", "Yes", "Yes", "No", "Yes", "No", "Yes"]
})

g = sns.catplot(x="Gender",
                y="Age", data=survey_data,
                kind="box", hue="Interested in Pets")

g.figure.suptitle("Age of Those Interested in Pets vs. Not")

plt.show()
```


Age of Those Interested in Pets vs. Not



```
In [42]: import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt

sns.set_style("dark")

survey_data = pd.DataFrame({
    "Village - town": ["Village", "Town", "Village", "Town", "Village", "Town"],
    "Likes Techno": [1, 0, 1, 1, 0, 1],
    "Gender": ["Male", "Female", "Female", "Male", "Male", "Female"]
})

g = sns.catplot(x="Village - town", y="Likes Techno",
                data=survey_data, kind="bar",
                col="Gender")

g.figure.suptitle("Percentage of Young People Who Like Techno", y=1.02)

g.set(xlabel="Location of Residence",
      ylabel="% Who Like Techno")

plt.show()
```

